Connections between extended Lambek calculus and Linear Logic

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Abstract

Lambek-Grishin logic (LG) is a substructural logic used in natural language processing. Several calculi have been proposed for LG to improve its proof-theoretic properties. Display sequent calculi exhibit a great modularity and a canonical cut-elimination theorem. Focused calculi force the implementation of a proof-search strategy thanks to additional side conditions on the applicability of inference rules. We introduce a novel sequent calculus **fD.LG** for LG which is both a display and a focused calculus. We present an algebraic semantics $\mathbb{FP}.\mathbb{LG}$ with respect to which **fD.LG** is sound and complete. The semantic footprint of focusing enables us to glance a possible refinement of connective behaviours, what may lead to a better understanding of substructurality and syntax-semantics compositionality of natural language.

Keywords: Lambek-Grishin calculus, Multi-type display calculi, Focused sequent calculi, Polarized logics, Heterogeneous algebras, Weakening relations, Semantics of proofs

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I Introduction

I did my Année de Recherche Prédoctorale à l'Étranger (ARPE) in Utrecht (the Netherlands). My supervisor was Michael Moortgat from the Utrecht Institute of Linguistics OTS (Universiteit Utrecht). I joined the project of his post-doc Giuseppe Greco, and began working closely with him in the second half of the internship.

The original topic (*Connections between extended Lambek grammars, linear λ-calculus and vector-based models of meaning*) was very broad. I did not touch to the last part because Michael found another post-doc to work on that. The mid-ARPE report betrays this lack of focused topic. It summarises the wide literature and gives a bunch of ideas and unstable models without strong theorem. However, it sets essential principles for the following study. In the second half, the tight collaboration with Giuseppe helped narrowing the approach, thus setting a more precise and reachable goal. The confrontation of our different viewpoints gave rise to a complex and novel system enjoying many interesting properties. The paper we wrote (submitted to CSL 2021) constitutes the main part of this report (section II).

The mid-ARPE report gives a general receipe to get a categorical and computational semantics for a large class of substructural logics by the mean of focusing and polarization. The studied logics are logics which can be expressed by a proper displayable sequent calculus [12]. This class of calculi enjoy sufficient conditions

for a canonical theorem of cut-elimination. One key condition is that every substructure of a sequent can be displayed as the sole precedent or (exclusively) succedent thanks to invertible rules. Focusing [1] is a proof search normalisation, originally developed for Linear Logic. The proof is divided between focused and non-focused phases depending on the polarity of connectives (positive = left adjoint and negative = right adjoint). The aim of focusing is to avoid irrelevant rule permutations to be able to define identity of sequent proofs, a problem in non-focused Gentzen calculus.

My initial goal to find a semantics where positive and negative operators are not interpreted identically therefore meets the project of Giuseppe to design a proper displayable calculus where every proof is focalized*. The main ingredient to obtain this system is to distinguish positive and negative formulas and structures as different sorts, and to interpret each sort in a different category / algebra. This was already present in the mid-ARPE report. Compared to the latter, the final article restricts itself to Lambek-Grishin calculus (LG) [23] ((co)residuated non-associative non-commutative and non-unital tensor $\otimes$ and cotensor $\oplus$) without distributivity, an algebraic semantics instead of a categorical one and no computational interpretation. Moreover, operational connectives and their structural counterparts are interpreted identically. The contributions of the article are the following: (i) a heterogeneous multi-type proper display calculus **fD.LG** for Lambek-Grishin logic (section 3.1) (ii) a theorem of strong focalization of every cut-free† derivation of **fD.LG** (theorem 23) (iii) a sound and complete semantics $\mathbb{FP}.\mathbb{LG}$ for **fD.LG** (sections 2.2 and 4).

The key idea to enforce strong focalization lies in the introduction of a second sort distinction: shifted formulas (i.e. beginning by a shift) vs. pure formula (i.e. beginning by a LG-connective or an atom). I defended that idea of mine to Giuseppe and it turned out to be successful. The design of the calculus was a common challenge of long and constant improvement. Once it was stable, I defined the semantics on a base of Giuseppe's ideas. I adapted the definition of strong focalization from Laurent [21] and wrote all the proofs of the article. I extended the class of multi-type proper display calculi to include calculi with heterogeneous sequents, and thus ours. By lack of space this theorem was not included in the article. I also developed a graphical tool (sequent multi-graphs) to visualise and reason topologically on sequent calculi. However, as it does not fit in the gist of the article, only a schematic simplification was included (Fig. 2). Appendix A contains a remark which was also left out, by lack of precise formulation, but which opens interesting considerations with respect to my initial topic.

II Article

*I use the word *focused* for phases and sequents but *focalized* for proofs, according to the literature.

†and ℓ, r-variant-free, see the aforesaid theorem.

Lambek-Grishin Calculus: Focusing, Display and Full Polarization

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Abstract

Focused sequent calculi are a refinement of sequent calculi, where additional side-conditions on the applicability of inference rules force the implementation of a proof search strategy. Focused cut-free proofs exhibit a special normal form that is used for defining identity of sequent calculi proofs. We introduce a novel focused display calculus **fD.LG** and a fully polarized algebraic semantics $\mathbb{FP}.\mathbb{LG}$ for Lambek-Grishin logic by generalizing the theory of *multi-type calculi* and their algebraic semantics with *heterogenous consequence relations*. We show that the calculus **fD.LG** has strong focalization and it is sound and complete w.r.t. $\mathbb{FP}.\mathbb{LG}$. We also prove that a Lambek-Grishin algebra can be equivalently presented as an $\mathbb{FP}.\mathbb{LG}$ algebra and, therefore, **fD.LG** is an alternative presentation of Lambek-Grishin logic.

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1 Introduction

The problem of the identity of proofs is a fundamental one. It has been actively investigated in philosophy and mathematics (when do two proofs correspond to the same argument?), and in computer sciences (when do two algorithms correspond to the same program?). A logic can be presented by different formalisms. Sequent calculi often exhibit syntactically different proofs of the very same end-sequent. Some of these proofs differ from each other by trivial permutations of inference rules. Other formalisms, like natural deduction calculi or proof nets, are less sensitive to inference rule permutations and are usually taken as benchmarks for defining identity of proofs. *Focused sequent calculi* [1, 2, 22] make use of syntactic restrictions on the applicability of inference rules achieving three main goals: (i) the proof search space is considerably reduced without losing completeness, (ii) every cut-free proof comes in a special normal form, (iii) leading to a criterion for defining identity of sequent calculi proofs. Being able to identify or tell apart two proofs has far-reaching consequences. In particular, in the tradition of parsing-as-deduction [19, 20], various logical systems – and notably various extensions of the Lambek calculus – have been proposed to recognise not only whether sentences are syntactically well-formed, but also to capture different semantic readings by 'genuinely different' proofs in the type logic [5, 24].

In this paper, we focus on the minimal Lambek-Grishin logic and we provide a novel algebraic and proof-theoretic analysis of the focused Lambek-Grishin calculus [24]. More in general, this analysis leads to the identification of a new class of display calculi and their natural algebraic semantics.



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The gist of the analysis is to generalise (and refine) *multi-type display calculi* and *heterogeneous algebras* ([7]) admitting not only heterogeneous operators, but also *heterogeneous consequence relations*, now interpreted as *weakening relations* [18] (i.e. a natural generalisation of partial orders). Here, we introduce a specific instance of this class tailored for the signature of the Lambek-Grishin logic and we plan to provide the general picture as future work. In particular, we plan to show that if a calculus belongs to this class, then it enjoys cut-elimination, aiming at generalizing the cut-elimination meta-theorem in the tradition of display calculi (see [26]). Moreover, we conjecture that any *displayable logic* ([13]) can be equivalently presented as an instance of this class. The next paragraph summarises the main features of this analysis in general terms, without special reference to Lambek-Grishin logic.

In the case of focused sequent calculi, the distinction between *positive* versus *negative* formulas is the key ingredient for organising proofs in so-called *phases*. The distinction is proof-theoretically relevant in that it reflects a fundamental distinction between logical introduction rules: the left introduction rules for positive connectives are *invertible* where the right introduction rules are *non-invertible* in general, and vice versa for negative connectives. We observe that this distinction is also semantically grounded, indeed positive formulas (in the original language of the logic) are left adjoints and negative formulas (in the original language of the logic) are right adjoints. Proofs in *focalized normal form* (see [24]) are cut-free proofs organised in three phases: two focused phases (either positive or negative) and non-focused phases. A focused positive (resp. negative) phase in a derivation is a sub-tree where a formula is decomposed as much as possible only by means of non-invertible logical rules for positive (resp. negative) connectives. This formula is said 'in focus'. All the other rules are applied only in non-focused phases. So, each derivable sequent has at most one formula in focus. Moreover, a non-focused phase always occur in between two focused phases.

So-called *shift operators* – usually denoted as $\uparrow$ and $\downarrow$ ([15, 16, 3]) – are often considered to polarize a focused sequent calculi, i.e. as a tool to control the interplay between positive and negative formulas and the interaction between phases. Shifts are adjoint unary operators that change the polarity of a formula, where $\uparrow$ goes from positive to negative, $\downarrow$ goes from negative to positive, and $\uparrow \dashv \downarrow$. In this paper, we consider positive and negative formulas as formulas of different sorts.[1] We also distinguish between positive (resp. negative) pure formulas and positive (resp. negative) shifted formulas, i.e. formulas under the scope of a shift operator, hence we call it a *full* polarization. So, we end up in considering four different sorts, each of which is interpreted in a different sub-algebra. Therefore, in this setting shifts are heterogeneous operators, where $\uparrow$ gets split into $\uparrow$ (from positive pure formulas into negative shifted formulas) and $\uparrow$ (from positive shifted formulas into negative pure formulas), $\downarrow$ gets split into $\downarrow$ (from negative pure formulas into positive shifted formulas) and $\downarrow$ (from negative shifted formulas into positive pure formulas), $\uparrow \dashv \downarrow$ and $\uparrow \dashv \downarrow$. Moreover, the composition of two shifts is still either a closure or an interior operator (by adjunction), but we do not assume that it is an identity.

The paper is structured as follows. In section 2 we introduce fully polarized LG-algebras $\mathbb{FP}.\mathbb{LG}$, and we show that Lambek-Grishin algebras $\mathbb{LG}$ can be equivalently presented in terms of $\mathbb{FP}.\mathbb{LG}$. In section 3 we introduce the focused display calculus **fD.LG** for the minimal Lambek-Grishin logic and we prove that it has the strong focalization property. In section 4, we show that the calculus **fD.LG** is sound and complete w.r.t. $\mathbb{FP}.\mathbb{LG}$.

[1] Note in the literature on multi-type display calculi 'types' is used instead of 'sorts'.

2 Algebraic semantics

In this section we first recall the definition of Lambek-Grishin algebras, weakening relations and their properties needed in what follows. Then, we define fully polarized LG-algebras. Finally, we show that Lambek-Grishin algebras can be equivalently presented as fully polarized LG-algebras (theorem 10).

2.1 Preliminaries

Lambek-Grishin algebras

The basic Lambek-Grishin logic **LG** [23] is the pure logic of residuation in the signature that expands the (non-unital, non-associative) Lambek calculus [20] with the so-called Grishin connectives (i.e. a co-tensor $\oplus$ and its residuals $\oslash, \odot$). **LG** is complete w.r.t. Lambek-Grishin algebras defined below.

▶ **Definition 1.** *A basic Lambek-Grishin algebra* $\mathbb{G} = (G, \leq, \otimes, \oplus, \setminus, \oslash, /, \odot)$ *is a partially ordered algebra endowed with six binary operations compatible with the order* $\leq$*. Moreover, the following residuation laws hold:*

$$B \leq A \setminus C \quad \textit{iff} \quad A \otimes B \leq C \quad \textit{iff} \quad A \leq C / B \qquad C \oslash B \leq A \quad \textit{iff} \quad C \leq A \oplus B \quad \textit{iff} \quad A \odot C \leq B \tag{1}$$

2.1.0.1 Weakening relations and collages

In this paper we use weakening relations [17, 18, 10, 11, 6] to interpret the heterogeneous consequence relations of the calculus introduced in section 3.1. Weakening relations can be viewed as the order-theoretic equivalents of profunctors [4] (aka distributors or bimodules), which have already been considered in models of polarized logic [15, 8]. In particular, partial orders are weakening relations where $\mathcal{A} = \mathcal{B}$ and $\leq_{\mathcal{A}} = \leq_{\mathcal{B}}$. We use $\preccurlyeq \subseteq \mathcal{A} \times \mathcal{B}$, $\preccurlyeq_{\mathcal{A}}^{\mathcal{B}}$ and $\mathcal{A} \nrightarrow \mathcal{B}$ interchangeably to denote a weakening relation with source $\mathcal{A}$ and target $\mathcal{B}$, and $\preccurlyeq_{\mathcal{A}}$ as an abbreviation for $\preccurlyeq_{\mathcal{A}}^{\mathcal{A}'}$. Given two relations R and S, we use RS to denote composition of relations.

▶ **Definition 2.** *A* ***weakening relation*** *is a relation* $\preccurlyeq \subseteq \mathcal{A} \times \mathcal{B}$ *on two partially ordered set* $(\mathcal{A}, \leq_{\mathcal{A}})$ *and* $(\mathcal{B}, \leq_{\mathcal{B}})$ *that is* compatible with the orders $\leq_{\mathcal{A}}$ *and* $\leq_{\mathcal{B}}$ *in the following sense*

$$\frac{A' \leq_{\mathcal{A}} A \qquad A \preccurlyeq B \qquad B \leq_{\mathcal{B}} B'}{A' \preccurlyeq B'}$$

▶ **Definition 3.** *Given two weakening relations* $\preccurlyeq_{\mathcal{A}} \subseteq \mathcal{A} \times \mathcal{A}'$ *and* $\preccurlyeq_{\mathcal{B}} \subseteq \mathcal{B} \times \mathcal{B}'$*, we say that the order-preserving functions* $L : \mathcal{A} \to \mathcal{B}$ *and* $R : \mathcal{B}' \to \mathcal{A}'$ *form a* ***heterogeneous adjoint pair*** $L \dashv_{\preccurlyeq_{\mathcal{A}}}^{\preccurlyeq_{\mathcal{B}}} R$ *if for every* $A \in \mathcal{A}$ *and* $B' \in \mathcal{B}'$*,*

$$L(A) \preccurlyeq_{\mathcal{B}} B' \textit{ iff } A \preccurlyeq_{\mathcal{A}} R(B') \qquad \begin{array}{ccc} \mathcal{A}' & \overset{R}{\longleftarrow} & \mathcal{B}' \\ {\scriptstyle \preccurlyeq_{\mathcal{A}}}\uparrow & \top & \uparrow{\scriptstyle \preccurlyeq_{\mathcal{B}}} \\ \mathcal{A} & \underset{L}{\longrightarrow} & \mathcal{B} \end{array} \tag{2}$$

If $\mathcal{A}' = \mathcal{A}$, $\preccurlyeq_{\mathcal{A}} = \leq_{\mathcal{A}}$, $\mathcal{B}' = \mathcal{B}$ and $\preccurlyeq_{\mathcal{B}} = \leq_{\mathcal{B}}$, we recover the usual definition of adjunction.

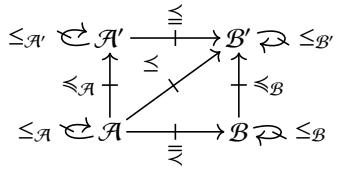
▶ **Proposition 4.** *If* $L \dashv_{\preccurlyeq_{\mathcal{A}}}^{\preccurlyeq_{\mathcal{B}}} R$ *is a heterogeneous adjunction, then it defines a weakening relation* $\preccurlyeq \subseteq \mathcal{A} \times \mathcal{B}'$ *by* $A \preccurlyeq B'$ *iff* $L(A) \preccurlyeq_{\mathcal{B}} B'$*, which is also equivalent to* $A \preccurlyeq_{\mathcal{A}} R(B')$*. We say that* $\preccurlyeq$ *is* ***the weakening relation represented by*** $L \dashv_{\preccurlyeq_{\mathcal{A}}}^{\preccurlyeq_{\mathcal{B}}} R$*.*

The proof only requires unfolding of the definitions (see appendix A).

▶ **Definition 5.** *If $\leqslant \subseteq \mathcal{A} \times \mathcal{B}$ is a weakening relation, then the relation $\leq_{\mathcal{A}\sqcup\mathcal{B}} := \leq_{\mathcal{A}} \sqcup \leqslant \sqcup \leq_{\mathcal{B}}$ defined on the disjoint union $\mathcal{A} \sqcup \mathcal{B}$ is an order. We call it the **collage order** on $\mathcal{A} \sqcup \mathcal{B}$.*

The collage order $(\mathcal{A} \sqcup \mathcal{B}, \leq_{\mathcal{A}\sqcup\mathcal{B}})$ corresponds to the collage [25] (or cograph) of $\leqslant$ seen as a profunctor. We extend the $\sqcup$ notation to weakening relations.

▶ **Definition 6.** *If we are in the following situation:*



*and we also have $\leqslant_{\mathcal{A}} \leqq \subseteq \leq$ and $\eqq \leqslant_{\mathcal{B}} \subseteq \leq$, then the relation $\overset{\circ}{\leq} := \leqq \sqcup \leq \sqcup \eqq$ is a weakening relation on the collage orders $\mathcal{A} \sqcup \mathcal{A}'$ and $\mathcal{B} \sqcup \mathcal{B}'$, and we call it the **collage weakening relation**.*

2.2 Fully polarized LG-algebra

We write $\mathcal{A}^\partial$ for the order dual of $(\mathcal{A}, \leq_{\mathcal{A}})$, i.e. $A \leq_{\mathcal{A}^\partial} A'$ iff $A' \leq_{\mathcal{A}} A$. We use P, Q (resp. $\dot{P}, \dot{Q}$) for pure (resp. shifted) positive elements, i.e. elements in the poset $\mathbb{P}$ (resp. in $\dot{\mathbb{P}}$); M, N (resp. $\dot{M}, \dot{N}$) for pure (resp. shifted) negative elements, i.e. elements in the poset $\mathbb{N}$ (resp. $\dot{\mathbb{N}}$); $\mathring{P}, \mathring{Q}, \mathring{R}$ (resp. $\mathring{M}, \mathring{N}, \mathring{L}$) for general positive (resp. negative) elements, i.e. elements in the poset $\mathring{\mathbb{P}}$ (resp. $\mathring{\mathbb{N}}$). The letters A, B, C are used whenever we do not need to specify the poset.

An *order-type* over $n \in \mathbb{N}$ is an n-tuple $\epsilon \in \{1, \partial\}^n$. For any order type ε, we let $\mathbb{A}^\varepsilon := \Pi_{i=1}^n \mathbb{A}^{\varepsilon_i}$. We use $n_h \in \mathbb{N}$ to denote the arity of a connective h. The language $\mathcal{L}_{\mathbb{FP}.\mathbb{LG}}(\mathcal{F}, \mathcal{G})$ (from now on abbreviated as $\mathcal{L}_{\mathbb{FP}.\mathbb{LG}}$) takes as parameters: two disjoint denumerable sets of proposition letters AtProp^+, elements of which are denoted p, q, and AtProp^-, elements of which are denoted m, n, and two disjoint sets of connectives:

$$\begin{array}{c} \mathcal{F} = \{\otimes, \otimes_\ell, \otimes_r, \oslash, \oslash_\ell, \oslash_r, \otimes, \otimes_\ell, \otimes_r, \uparrow, \mathfrak{1}\} \\ \mathcal{G} = \{\oplus, \oplus_\ell, \oplus_r, \backslash, \backslash_\ell, \backslash_r, /, /_\ell, /_r, \downarrow, \mathfrak{l}\} \end{array}$$

$$\begin{array}{c} \epsilon(\otimes) = \epsilon(\otimes_\ell) = \epsilon(\otimes_r) = \epsilon(\oplus) = \epsilon(\oplus_\ell) = \epsilon(\oplus_r) = (1,1) \\ \epsilon(\otimes) = \epsilon(\otimes_\ell) = \epsilon(\otimes_r) = \epsilon(\backslash) = \epsilon(\backslash_\ell) = \epsilon(\backslash_r) = (\partial,1) \\ \epsilon(\oslash) = \epsilon(\oslash_\ell) = \epsilon(\oslash_r) = \epsilon(/) = \epsilon(/_\ell) = \epsilon(/_r) = (1,\partial) \\ \epsilon(\downarrow) = \epsilon(\mathfrak{l}) = \epsilon(\uparrow) = \epsilon(\mathfrak{1}) = (1) \end{array} \tag{3}$$

▶ **Definition 7.** *A fully polarized LG-algebra ($\mathbb{FP}.\mathbb{LG}$) $\mathbb{A}$ is defined by four posets $(\mathbb{P}, \leq)$, $(\dot{\mathbb{P}}, \leqq)$, $(\mathbb{N}, \leq)$ and $(\dot{\mathbb{N}}, \leqq)$ together with*

- *Two adjunctions $\uparrow \dashv \downarrow$ and $\mathfrak{1} \dashv \mathfrak{l}$*



We use $\eqq$ for the weakening relation represented by $\uparrow \dashv \downarrow$ and $\leqq$ for the weakening relation represented by $\mathfrak{1} \dashv \mathfrak{l}$.

- *Three weakening relations $\eqq \subseteq \mathbb{P} \times \dot{\mathbb{P}}$, $\leq \subseteq \mathbb{P} \times \mathbb{N}$ and $\leqq \subseteq \dot{\mathbb{N}} \times \mathbb{N}$ such that for all $P \in \mathbb{P}$ and $N \in \mathbb{N}$ we have*

$$\uparrow P \leqq N \quad \textit{iff} \quad P \leq N \quad \textit{iff} \quad P \eqq \downarrow N \tag{5}$$

i.e. $\preceq$ *is the weakening relation represented by the heterogeneous adjunction* $\uparrow \dashv \substack{\leqq \\ \leq} \downarrow$.
We define the collage posets $(\mathring{\mathbb{P}}, \mathring{\leq}) = (\mathbb{P} \sqcup \dot{\mathbb{P}}, \leq \sqcup \dot{\leqq} \sqcup \dot{\leqq})$, $(\mathring{\mathbb{N}}, \mathring{\leq}) = (\mathbb{N} \sqcup \dot{\mathbb{N}}, \leq \sqcup \dot{\leqq} \sqcup \dot{\leqq})$ *and the collage weakening relation* $\mathring{\leq} = \dot{\leqq} \sqcup \preceq \sqcup \dot{\leqq} \subseteq \mathring{\mathbb{P}} \times \mathring{\mathbb{N}}$, *summarised in Fig. 1.*

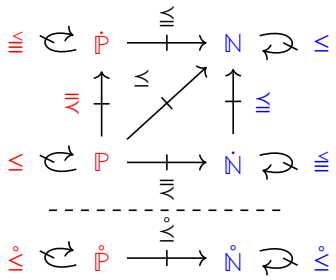


Figure 1 Weakening relations in $\mathbb{FP}.\mathbb{LG}$-algebras.

- *Six operations (that we call LG-connectives)*

$$\begin{array}{llllll} \otimes: & \mathring{\mathbb{P}} \times \mathring{\mathbb{P}} \to \mathbb{P} & \oslash: & \mathring{\mathbb{P}} \times \mathring{\mathbb{N}}^{\partial} \to \mathbb{P} & \oslash: & \mathring{\mathbb{N}}^{\partial} \times \mathring{\mathbb{P}} \to \mathbb{P} \\ \oplus: & \mathring{\mathbb{N}} \times \mathring{\mathbb{N}} \to \mathbb{N} & \backslash: & \mathring{\mathbb{P}}^{\partial} \times \mathring{\mathbb{N}} \to \mathbb{N} & / : & \mathring{\mathbb{N}} \times \mathring{\mathbb{P}}^{\partial} \to \mathbb{N} \end{array}$$

such that the following heterogeneous adjunctions hold

$$\begin{array}{llllll} \mathring{Q} \mathring{\leq} \mathring{P} \backslash \mathring{N} & \textit{iff} & \mathring{P} \otimes \mathring{Q} \mathring{\leq} \mathring{N} & \textit{iff} & \mathring{P} \mathring{\leq} \mathring{N} / \mathring{Q} \\ \mathring{P} \oslash \mathring{N} \mathring{\leq} \mathring{M} & \textit{iff} & \mathring{P} \mathring{\leq} \mathring{M} \oplus \mathring{N} & \textit{iff} & \mathring{M} \oslash \mathring{P} \mathring{\leq} \mathring{N} \end{array} \tag{6}$$

- *Finally, 12 operations (that we call ℓ-variants and r-variants, or simply LG-variants)*

$$\begin{array}{llllll} \otimes_\ell: & \mathring{\mathbb{N}} \times \mathring{\mathbb{P}} \to \dot{\mathbb{N}} & \oslash_\ell: & \mathring{\mathbb{N}} \times \mathring{\mathbb{N}}^{\partial} \to \dot{\mathbb{N}} & \oslash_\ell: & \mathring{\mathbb{P}}^{\partial} \times \mathring{\mathbb{P}} \to \dot{\mathbb{N}} \\ \oplus_\ell: & \mathring{\mathbb{P}} \times \mathring{\mathbb{N}} \to \dot{\mathbb{P}} & \backslash_\ell: & \mathring{\mathbb{N}}^{\partial} \times \mathring{\mathbb{N}} \to \dot{\mathbb{P}} & /_\ell: & \mathring{\mathbb{P}} \times \mathring{\mathbb{P}}^{\partial} \to \dot{\mathbb{P}} \\ \otimes_r: & \mathring{\mathbb{P}} \times \mathring{\mathbb{N}} \to \dot{\mathbb{N}} & \oslash_r: & \mathring{\mathbb{P}} \times \mathring{\mathbb{P}}^{\partial} \to \dot{\mathbb{N}} & \oslash_r: & \mathring{\mathbb{N}}^{\partial} \times \mathring{\mathbb{N}} \to \dot{\mathbb{N}} \\ \oplus_r: & \mathring{\mathbb{N}} \times \mathring{\mathbb{P}} \to \dot{\mathbb{P}} & \backslash_r: & \mathring{\mathbb{P}}^{\partial} \times \mathring{\mathbb{P}} \to \dot{\mathbb{P}} & /_r: & \mathring{\mathbb{N}} \times \mathring{\mathbb{N}}^{\partial} \to \dot{\mathbb{P}} \end{array}$$

such that the following adjunctions hold

$$\begin{array}{llllllllllll} \mathring{Q} \mathring{\leq} \mathring{P} \backslash_r \mathring{R} & \textit{iff} & \mathring{P} \otimes \mathring{Q} \mathring{\leq} \mathring{R} & \textit{iff} & \mathring{P} \mathring{\leq} \mathring{R} /_\ell \mathring{Q}, & \mathring{L} \oslash_\ell \mathring{N} \mathring{\leq} \mathring{M} & \textit{iff} & \mathring{L} \mathring{\leq} \mathring{M} \oplus \mathring{N} & \textit{iff} & \mathring{M} \oslash_r \mathring{L} \mathring{\leq} \mathring{N} \\ \mathring{Q} \mathring{\leq} \mathring{L} \backslash_\ell \mathring{N} & \textit{iff} & \mathring{L} \otimes_\ell \mathring{Q} \mathring{\leq} \mathring{N} & \textit{iff} & \mathring{L} \mathring{\leq} \mathring{N} / \mathring{Q}, & \mathring{P} \oslash_r \mathring{R} \mathring{\leq} \mathring{M} & \textit{iff} & \mathring{P} \mathring{\leq} \mathring{M} \oplus_r \mathring{R} & \textit{iff} & \mathring{M} \oslash \mathring{P} \mathring{\leq} \mathring{R} \\ \mathring{Q} \mathring{\leq} \mathring{P} \backslash \mathring{N} & \textit{iff} & \mathring{P} \otimes_r \mathring{L} \mathring{\leq} \mathring{N} & \textit{iff} & \mathring{P} \mathring{\leq} \mathring{N} /_r \mathring{L}, & \mathring{P} \oslash \mathring{N} \mathring{\leq} \mathring{R} & \textit{iff} & \mathring{P} \mathring{\leq} \mathring{R} \oplus_\ell \mathring{N} & \textit{iff} & \mathring{R} \oslash_\ell \mathring{P} \mathring{\leq} \mathring{N} \end{array} \tag{7}$$

▶ **Proposition 8.** *In any* $\mathbb{FP}.\mathbb{LG}$ *we have* $\dot{\leq} \dot{\leqq} = \preceq = \dot{\leqq} \dot{\leqq}$.

Proof. We show that $\dot{\leq} \dot{\leqq} = \preceq$. Fix $P \in \mathbb{P}$, $N \in \mathbb{N}$ and assume that $P \dot{\leq} \dot{Q}$ and $\dot{Q} \dot{\leqq} N$ for some $\dot{Q} \in \dot{\mathbb{P}}$. From $\dot{Q} \dot{\leqq} N$, we conclude that $\dot{Q} \dot{\leqq} \downarrow N$, for $\dot{\leqq}$ is the weakening relation represented by $\uparrow \dashv \downarrow$ (see proposition 4). From $P \dot{\leq} \dot{Q}$ and $\dot{Q} \dot{\leqq} \downarrow N$ we conclude $P \dot{\leq} \downarrow N$, for $\dot{\leq}$ is a weakening relation compatible with the partial order $\dot{\leqq}$. Therefore, $P \preceq N$ by (5).

Now, fix $P \in \mathbb{P}$, $N \in \mathbb{N}$ and assume $P \preceq N$. On the one hand, $P \dot{\leq} \downarrow N$ by (5). On the other hand, $\downarrow N \dot{\leqq} \downarrow N$ gives $\downarrow N \dot{\leqq} N$ by proposition 4 on $\dot{\leqq}$. The equality $\preceq = \dot{\leq} \dot{\leqq}$ is proven in a similar way. ◀

▶ Remark 9. In [18], operations that are order-reversing in some coordinate are considered 'problematic', essentially because source and target of weakening relations are, in general, of different types. Remark 2.23 in [18] illustrates the concern considering negation and implication in Boolean or Heyting algebras as prototypical examples. We observe that this is an issue only insofar we confine ourselves to homogeneous operations. In the present setting, the problem is overcome allowing heterogeneous operations. In the case of fully polarized LG-algebras, shifts, the adjoints of shifts, LG-connectives, and LG-variants are heterogeneous operations (see definition 7).

2.3 $\mathbb{FP}.\mathbb{LG}$ as equivalent presentations of $\mathbb{LG}$

▶ **Theorem 10.** *Every LG-algebra can be equivalently presented as a $\mathbb{FP}.\mathbb{LG}$ algebra.*

Proof. ($\Rightarrow$) Fix a LG-algebra $\mathbb{G} = (G, \leq, \otimes^{\mathbb{G}}, /^{\mathbb{G}}, \setminus^{\mathbb{G}}, \oplus^{\mathbb{G}}, \oslash^{\mathbb{G}}, \otimes^{\mathbb{G}})$. We construct a $\mathbb{FP}.\mathbb{LG}$-algebra $\mathbb{A}$ by taking a copy of G as the domain of any sub-algebra in $\mathbb{A}$, by defining shifts as maps sending an element to its copy in the appropriate sub-algebra of $\mathbb{A}$, and finally, for each A, B in the appropriate sub-algebra of $\mathbb{A}$, for each weakening relation R in $\mathbb{A}$, and for each binary operation $\star^{\mathbb{A}}$ in $\mathbb{A}$, by defining

$$A \, R \, B \;\text{ iff }\; A \leq B \quad \text{and} \quad A \star^{\mathbb{A}} B \;\text{ iff }\; A \star^{\mathbb{G}} B$$

It is straightforward to check that weakening relations and operations are well-defined, and (5), (6), and (7) hold.

($\Leftarrow$) Fix a $\mathbb{FP}.\mathbb{LG}$-algebra $\mathbb{A}$. Recall that in any $\mathbb{FP}.\mathbb{LG}$-algebra the poset $\mathring{\mathbb{P}}$ (resp. $\mathring{\mathbb{N}}$) is the disjoint union of positive (resp. negative) pure elements and positive (resp. negative) shifted elements. We adopt the following convention

$$\mathring{\mathbb{P}} \ni A^+ = \begin{cases} A & \text{if } A \in \mathring{\mathbb{P}} \\ \downarrow A & \text{if } A \in \mathbb{N} \\ \downarrow A & \text{if } A \in \dot{\mathbb{N}} \end{cases} \quad \text{and} \quad \mathring{\mathbb{N}} \ni A^- = \begin{cases} A & \text{if } A \in \mathring{\mathbb{N}} \\ \uparrow A & \text{if } A \in \mathbb{P} \\ \uparrow A & \text{if } A \in \dot{\mathbb{P}} \end{cases} \tag{8}$$

We construct a LG-algebra $\mathbb{G} = (G, \leq, \otimes^{\mathbb{G}}, /^{\mathbb{G}}, \setminus^{\mathbb{G}}, \oplus^{\mathbb{G}}, \oslash^{\mathbb{G}}, \otimes^{\mathbb{G}})$ by taking $G = \mathbb{P} \sqcup \dot{\mathbb{P}} \sqcup \mathbb{N} \sqcup \dot{\mathbb{N}}$, by defining $A \leq B$ iff $A^+ \mathrel{\mathring{\leq}} B^-$, and by defining the operations as follows

$$\begin{array}{lll} A \otimes^{\mathbb{G}} B := A^+ \otimes^{\mathbb{A}} B^+ & A \oslash^{\mathbb{G}} B := A^+ \oslash^{\mathbb{A}} B^- & A \otimes^{\mathbb{G}} B := A^- \otimes^{\mathbb{A}} B^+ \\ A \oplus^{\mathbb{G}} B := A^- \oplus^{\mathbb{A}} B^- & A \setminus^{\mathbb{G}} B := A^+ \setminus^{\mathbb{A}} B^- & A /^{\mathbb{G}} B := A^- /^{\mathbb{A}} B^+ \end{array} \tag{9}$$

It is straightforward to check the partial order $\leq$ and operations are well-defined, and that indeed $\mathbb{G}$ is a LG-algebra accordingly to definition 1. For instance, below we show that $P \otimes^{\mathbb{G}} N$ is residuated in its second coordinate (see (1)) assuming that $A \in \mathbb{P}$, $B \in \mathbb{N}$ and $C \in \dot{\mathbb{P}}$:

$$\begin{array}{rl} P \otimes^{\mathbb{G}} N \leq \dot{Q} \text{ iff} & P \otimes^{\mathbb{A}} \downarrow N \mathrel{\mathring{\leq}} \uparrow \dot{Q} \\ \text{iff} & P \mathrel{\mathring{\leq}} \uparrow \dot{Q} /^{\mathbb{A}} \downarrow N \\ \text{iff} & P \leq \dot{Q} /^{\mathbb{G}} N \end{array}$$

◀

3 Proof theory

The basic Lambek-Grishin logic can be presented as a (single-type) proper display sequent calculus (see [23]). Section 2.1 of [24] provides a display sequent calculus for the basic Lambek-Grishin logic and its expansion with Grishin's [14] 'linear distributivity' structural rules capturing interaction between the $\otimes$ and the $\oplus$ families of connectives. Section 3.1 of [24] provides a focused sequent calculus **f.LG** for the same logic. In [24] the calculus **f.LG** is considered a display calculus given that all the connectives in the language are residuated in each coordinate, even tough the so-called *display postulates* capturing residuation can only be applied in neutral phases. On the contrary, all the connectives of the calculus **fD.LG** introduced in section 3.1 are residuated and display postulates can be applied *in any phase*. Therefore, **fD.LG** is a display calculus accordingly to the usual definition. Even though we will not expand on this point in what follows, it is worth to mention that **f.LG** and **fD.LG** are equivalent calculi. Indeed, it is not difficult to define faithful translations from **f.LG**-derivations to **fD.LG**-derivations and vice versa.

More in general, **fD.LG** has the following distinctive features: (i) homogeneous as well as heterogeneous connectives are considered (multi-type), (ii) each rule is closed under uniform substitution within each type (properness), (iii) every structure occurring in a derivable sequent can be isolated either in precedent or, exclusively, in succedent position by means of display postulates (display property), and (iv) homogeneous as well as heterogeneous turnstiles are considered. Any *multi-type proper display calculus* (see [26, 9]) has features i-iii but not iv.

3.1 Focused display LG-calculus

The language of **fD.LG** is the Lambek-Grishin display calculus language expanded with the structural ℓ, r-variants and shifts operators.

▶ **Notation 11.** *Following the display calculi literature [12], we adopt a notation where structural and operational (aka logical) connectives are in a one-to-one correspondence. Moreover, we mark the structural counterpart of a connective $\star$ as follows: $\hat{\star}$ if $\star$ is a left-adjoint or residual, $\check{\star}$ if $\star$ is a right-adjoint or residual.*

The Lambek-Grishin structural and operational connectives are the following

Structural symbols	$\hat{\otimes}$	$\hat{\oslash}$	$\hat{\otimes}$	$\check{\oplus}$	$\check{\backslash}$	$\check{/}$
Operational symbols	$\otimes$	$\oslash$	$\otimes$	$\oplus$	$\backslash$	$/$

Below we list the structural ℓ, r-variants included in the language of **fD.LG** and the corresponding operational ℓ, r-variants (in grey cells). We consider ℓ, r-variants essentially to ensure the display property, therefore we find more convenient to not include at all the operational ℓ, r-variants in the language of **fD.LG**. The subscript ℓ (resp. r) of a ℓ-variant $\star_\ell$ (resp. r-variant $\star_r$) indicates that the subformula on its left is of the opposite polarity w.r.t. the corresponding LG-connective $\star$.

Structural symbols	$\hat{\otimes}_\ell$	$\hat{\oslash}_\ell$	$\hat{\otimes}_\ell$	$\check{\oplus}_\ell$	$\check{\backslash}_\ell$	$\check{/}_\ell$	$\hat{\otimes}_r$	$\hat{\oslash}_r$	$\hat{\otimes}_r$	$\check{\oplus}_r$	$\check{\backslash}_r$	$\check{/}_r$
Operational symbols	$\otimes_\ell$	$\oslash_\ell$	$\otimes_\ell$	$\oplus_\ell$	$\backslash_\ell$	$/_\ell$	$\otimes_r$	$\oslash_r$	$\otimes_r$	$\oplus_r$	$\backslash_r$	$/_r$

Below we list the structural and operational shifts operators. We find more convenient to not include the operational adjoints of shifts (in grey cells) in the language of **fD.LG**

Structural symbols	$\check{\downarrow}$	$\check{\downarrow}$	$\hat{\uparrow}$	$\hat{\uparrow}$
Operational symbols	$\downarrow$	$\downarrow$	$\uparrow$	$\uparrow$

For any connective h (either structural or operational), the arity n_h, order-type $\epsilon(h)$, and its classification as $\mathcal{F}$-connective or , exclusively, $\mathcal{G}$-connective are like in 3.

▶ **Notation 12.** *We adopt the following notational convention for formulas and structures: $\mathring{P} \in \{P, \dot{P}\}, \mathring{X} \in \{X, \dot{X}\}, \mathring{N} \in \{N, \dot{N}\}, \mathring{\Delta} \in \{\Delta, \dot{\Delta}\}$. For instance, accordingly to this convention, we have that $\mathring{P} \otimes \mathring{Q} \in \{P \otimes Q, \dot{P} \otimes Q, P \otimes \dot{Q}, \dot{P} \otimes \dot{Q}\}$. Therefore, general formulas and structures are not a full-fledged sort, but rather an abbreviation.*

The calculus **fD.LG** manipulates formulas and structures defined by the following mutual recursion, where $p \in \mathsf{AtProp}^+$ and $n \in \mathsf{AtProp}^-$:

$$
\begin{array}{rcll}
\mathsf{PurePosFm} \ni P & ::= & p \mid \mathring{P} \otimes \mathring{P} \mid \mathring{P} \oslash \mathring{N} \mid \mathring{N} \osl \mathring{P} & \text{Pure positive formulas} \\
\mathsf{PureNegFm} \ni N & ::= & n \mid \mathring{N} \oplus \mathring{N} \mid \mathring{P} \setminus \mathring{N} \mid \mathring{N} / \mathring{P} & \text{Pure negative formulas} \\
\mathsf{ShiftedPosFm} \ni \dot{P} & ::= & \downarrow N & \text{Shifted positive formulas} \\
\mathsf{ShiftedNegFm} \ni \dot{N} & ::= & \uparrow P & \text{Shifted negative formulas} \\
\mathsf{GenPosFm} \ni \mathring{P} & ::= & P \mid \dot{P} & \text{General positive formulas} \\
\mathsf{GenNegFm} \ni \mathring{N} & ::= & N \mid \dot{N} & \text{General negative formulas} \\
\mathsf{PurPosStr} \ni X & ::= & P \mid \dot{\downarrow} \dot{\Delta} \mid \mathring{X} \hat{\otimes} \mathring{X} \mid \mathring{X} \hat{\oslash} \mathring{\Delta} \mid \mathring{\Delta} \hat{\osl} \mathring{X} & \text{Pure positive structures} \\
\mathsf{PurNegStr} \ni \Delta & ::= & N \mid \hat{\uparrow} \dot{X} \mid \mathring{\Delta} \check{\oplus} \mathring{\Delta} \mid \mathring{X} \check{\setminus} \mathring{\Delta} \mid \mathring{\Delta} \check{/} \mathring{X} & \text{Pure negative structures} \\
\mathsf{ShiftedPosStr} \ni \dot{X} & ::= & \dot{P} \mid \dot{\downarrow} \Delta \mid \mathring{X} \check{\oplus}_\ell \mathring{\Delta} \mid \mathring{\Delta} \check{\oplus}_r \mathring{X} \mid & \text{Shifted positive structures} \\
 & & \mathring{\Delta} \check{\setminus}_\ell \mathring{\Delta} \mid \mathring{X} \check{\setminus}_r \mathring{X} \mid \mathring{X} \check{/}_\ell \mathring{X} \mid \mathring{\Delta} \check{/}_r \mathring{\Delta} & \\
\mathsf{ShiftedNegStr} \ni \dot{\Delta} & ::= & \dot{N} \mid \hat{\uparrow} X \mid \mathring{\Delta} \hat{\otimes}_\ell \mathring{X} \mid \mathring{X} \hat{\otimes}_r \mathring{\Delta} \mid & \text{Shifted negative structures} \\
 & & \mathring{\Delta} \hat{\oslash}_\ell \mathring{\Delta} \mid \mathring{X} \hat{\oslash}_r \mathring{X} \mid \mathring{X} \hat{\osl}_\ell \mathring{X} \mid \mathring{\Delta} \hat{\osl}_r \mathring{\Delta} & \\
\mathsf{GenPosStr} \ni \mathring{X} & ::= & X \mid \dot{X} & \text{General positive structures} \\
\mathsf{GenNegStr} \ni \mathring{\Delta} & ::= & \Delta \mid \dot{\Delta} & \text{General negative structures}
\end{array}
$$

The well-formed sequents are the following:

Positive sequents	$X \vdash Y$	$\dot{X} \Vdash Y$	$X \nVdash \dot{Y}$	$\dot{X} \Vvdash \dot{Y}$
Negative sequents	$\Delta \vdash \Gamma$	$\dot{\Delta} \Vdash \Gamma$	$\Delta \nVdash \dot{\Gamma}$	$\dot{\Delta} \Vvdash \dot{\Gamma}$
Neutral sequents	$X \vdash \Delta$	$\dot{X} \Vdash \Delta$	$X \nVdash \dot{\Delta}$	$\dot{X} \Vvdash \dot{\Delta}$

(10)

▶ **Notation 13.** *We extend the previous conventions to sequents as follows:* $\mathring{\vdash} \in \{\vdash, \Vdash, \nVdash, \Vvdash\}$, $\mathring{\vdash} \in \{\vdash, \Vdash, \nVdash, \Vvdash\}$, $\mathring{\vdash} \in \{\vdash, \Vdash, \nVdash, \Vvdash\}$. *The reading is supposed to preserve well-formedness. For instance, in a premise of a binary logical rule* $\mathring{\vdash} = \vdash$ *iff* $\mathring{X} = X$ *and* $\mathring{Y} = Y$, *or* $\mathring{\vdash} = \nVdash$ *iff* $\mathring{X} = X$ *and* $\mathring{Y} = \dot{Y}$, *and so on. Therefore, each binary logical rule below denotes four different rules. Nonetheless, notice that in any actual derivation the instantiation of a logical inference rule is unique and completely deterministic.*

The calculus **fD.LG** consists of the following rules.

Axioms and cuts

$$
\frac{}{p \vdash p}\ p\text{-Id} \qquad n\text{-Id}\ \frac{}{n \vdash n}
$$

$$
\text{P-Cut}\ \frac{\mathring{X} \mathring{\vdash} \mathring{P} \qquad \mathring{P} \mathring{\vdash} \mathring{Y}}{\mathring{X} \mathring{\vdash} \mathring{Y}} \qquad \frac{\mathring{\Gamma} \mathring{\vdash} \mathring{N} \qquad \mathring{N} \mathring{\vdash} \mathring{\Delta}}{\mathring{\Gamma} \mathring{\vdash} \mathring{\Delta}}\ \text{N-Cut}
$$

$$
\text{Pn-Cut}\ \frac{\mathring{X} \mathring{\vdash} \mathring{P} \qquad \mathring{P} \mathring{\vdash} \mathring{\Delta}}{\mathring{X} \mathring{\vdash} \mathring{\Delta}} \qquad \frac{\mathring{X} \mathring{\vdash} \mathring{N} \qquad \mathring{N} \mathring{\vdash} \mathring{\Delta}}{\mathring{X} \mathring{\vdash} \mathring{\Delta}}\ \text{nN-Cut}
$$

(11)

Logical rules

The logical rules transforming a structural connective in the premise into its logical counterpart in the conclusion are called *translation rules*. All the other logical rules are called *tonicity rules*. In the literature on focused calculi, 'asynchronous' and 'synchronous', respectively, are often used (e.g. in

[2]).

$$\otimes_L \frac{\mathring{P} \hat{\otimes} \mathring{Q} \overset{\circ}{\vdash} \mathring{\Delta}}{\mathring{P} \otimes \mathring{Q} \overset{\circ}{\vdash} \mathring{\Delta}} \quad \frac{\mathring{X} \overset{\circ}{\vdash} \mathring{P} \qquad \mathring{Y} \overset{\circ}{\vdash} \mathring{Q}}{\mathring{X} \hat{\otimes} \mathring{Y} \vdash \mathring{P} \otimes \mathring{Q}} \otimes_R \qquad \oplus_L \frac{\mathring{N} \overset{\circ}{\vdash} \mathring{\Gamma} \qquad \mathring{M} \overset{\circ}{\vdash} \mathring{\Delta}}{\mathring{N} \oplus \mathring{M} \vdash \mathring{\Gamma} \check{\oplus} \mathring{\Delta}} \quad \frac{\mathring{X} \overset{\circ}{\vdash} \mathring{N} \check{\oplus} \mathring{M}}{\mathring{X} \overset{\circ}{\vdash} \mathring{N} \oplus \mathring{M}} \oplus_R$$

$$\oslash_L \frac{\mathring{P} \hat{\oslash} \mathring{N} \overset{\circ}{\vdash} \mathring{\Delta}}{\mathring{P} \oslash \mathring{N} \overset{\circ}{\vdash} \mathring{\Delta}} \quad \frac{\mathring{X} \overset{\circ}{\vdash} \mathring{P} \qquad \mathring{N} \overset{\circ}{\vdash} \mathring{\Delta}}{\mathring{X} \hat{\oslash} \mathring{\Delta} \vdash \mathring{P} \oslash \mathring{N}} \oslash_R \qquad \backslash_L \frac{\mathring{X} \overset{\circ}{\vdash} \mathring{P} \qquad \mathring{N} \overset{\circ}{\vdash} \mathring{\Delta}}{\mathring{P} \backslash \mathring{N} \vdash X \check{\backslash} \Delta} \quad \frac{\mathring{X} \overset{\circ}{\vdash} \mathring{P} \check{\backslash} \mathring{N}}{\mathring{X} \overset{\circ}{\vdash} \mathring{P} \backslash \mathring{N}} \backslash_R$$

$$\oslash_L \frac{\mathring{N} \hat{\oslash} \mathring{P} \overset{\circ}{\vdash} \mathring{\Delta}}{\mathring{N} \oslash \mathring{P} \overset{\circ}{\vdash} \mathring{\Delta}} \quad \frac{\mathring{N} \overset{\circ}{\vdash} \mathring{\Delta} \qquad \mathring{X} \overset{\circ}{\vdash} \mathring{P}}{\mathring{\Delta} \hat{\oslash} \mathring{X} \vdash \mathring{N} \oslash \mathring{P}} \oslash_R \qquad / _L \frac{\mathring{N} \overset{\circ}{\vdash} \mathring{\Delta} \qquad \mathring{X} \overset{\circ}{\vdash} \mathring{P}}{\mathring{N} / \mathring{P} \vdash \mathring{\Delta} \check{/} \mathring{X}} \quad \frac{\mathring{X} \overset{\circ}{\vdash} \mathring{N} \check{/} \mathring{P}}{\mathring{X} \overset{\circ}{\vdash} \mathring{N} / \mathring{P}} / _R \tag{12}$$

$$\downarrow_L \frac{N \vdash \Delta}{\downarrow N \Vdash \check{\downarrow} \Delta} \quad \frac{\mathring{X} \overset{\circ}{\vdash} \check{\downarrow} N}{\mathring{X} \overset{\circ}{\vdash} \downarrow N} \downarrow_R \qquad \uparrow_L \frac{\hat{\uparrow} P \overset{\circ}{\vdash} \mathring{\Delta}}{\uparrow P \overset{\circ}{\vdash} \mathring{\Delta}} \quad \frac{X \vdash P}{\hat{\uparrow} X \Vdash \uparrow P} \uparrow_R$$

Display postulates

Below we use a double inference line to denote two rules: (i) from the premise to the conclusion and (ii) from the conclusion to the premise. We use the same name for both rules.

$$\begin{array}{ll} \hat{\otimes} \dashv \check{\backslash} \dfrac{\mathring{Y} \overset{\circ}{\vdash} \mathring{X} \check{\backslash} \mathring{\Delta}}{\mathring{X} \hat{\otimes} \mathring{Y} \overset{\circ}{\vdash} \mathring{\Delta}} \quad \dfrac{\mathring{Y} \overset{\circ}{\vdash} \mathring{X} \check{\backslash}_r \mathring{Z}}{\mathring{X} \hat{\otimes} \mathring{Y} \overset{\circ}{\vdash} \mathring{Z}} \hat{\otimes} \dashv \check{\backslash}_r & \hat{\oslash}_\ell \dashv \check{\oplus} \dfrac{\mathring{\Sigma} \hat{\oslash}_\ell \mathring{\Delta} \overset{\circ}{\vdash} \mathring{\Gamma}}{\mathring{\Sigma} \overset{\circ}{\vdash} \mathring{\Gamma} \check{\oplus} \mathring{\Delta}} \quad \dfrac{\mathring{X} \hat{\oslash} \mathring{\Delta} \overset{\circ}{\vdash} \mathring{\Gamma}}{\mathring{X} \overset{\circ}{\vdash} \mathring{\Gamma} \check{\oplus} \mathring{\Delta}} \hat{\oslash} \dashv \check{\oplus} \\ \hat{\otimes} \dashv \check{/} \dfrac{\mathring{X} \overset{\circ}{\vdash} \mathring{\Delta} \check{/} \mathring{Y}}{\mathring{X} \hat{\otimes} \mathring{Y} \overset{\circ}{\vdash} \mathring{\Delta}} \quad \dfrac{\mathring{X} \overset{\circ}{\vdash} \mathring{Z} \check{/}_\ell \mathring{Y}}{\mathring{X} \hat{\otimes} \mathring{Y} \overset{\circ}{\vdash} \mathring{Z}} \hat{\otimes} \dashv \check{\backslash}_\ell & \hat{\oslash}_r \dashv \check{\oplus} \dfrac{\mathring{\Gamma} \overset{\circ}{\vdash} \mathring{\Delta} \check{\oplus}_r \mathring{\Sigma}}{\mathring{\Gamma} \hat{\oslash}_r \mathring{\Sigma} \overset{\circ}{\vdash} \mathring{\Delta}} \quad \dfrac{\mathring{X} \hat{\oslash} \mathring{\Delta} \overset{\circ}{\vdash} \mathring{\Gamma}}{\mathring{X} \overset{\circ}{\vdash} \mathring{\Gamma} \check{\oplus} \mathring{\Delta}} \hat{\oslash} \dashv \check{\oplus} \end{array}$$

$$\begin{array}{ll} \hat{\oslash} \dashv \check{\oplus}_r \dfrac{\mathring{\Gamma} \hat{\oslash} \mathring{X} \overset{\circ}{\vdash} \mathring{Y}}{\mathring{X} \overset{\circ}{\vdash} \mathring{\Gamma} \check{\oplus}_r \mathring{Y}} \quad \dfrac{\mathring{X} \hat{\oslash} \mathring{\Delta} \overset{\circ}{\vdash} \mathring{Y}}{\mathring{X} \overset{\circ}{\vdash} \mathring{Y} \check{\oplus}_\ell \mathring{\Delta}} \hat{\oslash} \dashv \check{\oplus}_\ell & \hat{\otimes}_r \dashv \check{\backslash} \dfrac{\mathring{\Gamma} \overset{\circ}{\vdash} \mathring{X} \check{\backslash} \mathring{\Delta}}{\mathring{X} \hat{\otimes}_r \mathring{\Gamma} \overset{\circ}{\vdash} \mathring{\Delta}} \quad \dfrac{\mathring{Y} \overset{\circ}{\vdash} \mathring{\Gamma} \check{\backslash}_\ell \mathring{\Delta}}{\mathring{\Gamma} \hat{\otimes}_\ell \mathring{Y} \overset{\circ}{\vdash} \mathring{\Delta}} \hat{\otimes}_\ell \dashv \check{\backslash}_\ell \\ \hat{\oslash}_r \dashv \check{\oplus}_r \dfrac{\mathring{X} \overset{\circ}{\vdash} \mathring{\Gamma} \check{\oplus}_r \mathring{Y}}{\mathring{X} \hat{\oslash}_r \mathring{Y} \overset{\circ}{\vdash} \mathring{\Gamma}} \quad \dfrac{\mathring{Y} \overset{\circ}{\vdash} \mathring{Y} \check{\oplus}_\ell \mathring{\Delta}}{\mathring{Y} \hat{\oslash}_\ell \mathring{X} \overset{\circ}{\vdash} \mathring{\Delta}} \hat{\oslash}_\ell \dashv \check{\oplus}_\ell & \hat{\otimes}_r \dashv \check{/} \dfrac{\mathring{\Gamma} \overset{\circ}{\vdash} \mathring{\Delta} \check{/}_r \mathring{\Gamma}}{\mathring{X} \overset{\circ}{\vdash} \mathring{\Delta} \check{/}_r \mathring{\Gamma}} \quad \dfrac{\mathring{Y} \overset{\circ}{\vdash} \mathring{\Gamma} \check{\backslash}_\ell \mathring{\Delta}}{\mathring{\Gamma} \overset{\circ}{\vdash} \mathring{\Delta} \check{/} \mathring{Y}} \hat{\otimes}_\ell \dashv \check{/} \end{array} \tag{13}$$

$$\frac{\hat{\uparrow} X \Vdash \dot{\Delta}}{X \vdash \check{\downarrow} \dot{\Delta}} \hat{\uparrow} \dashv \check{\downarrow} \qquad \frac{\hat{\uparrow} X \Vdash \Delta}{X \Vdash \check{\downarrow} \Delta} \hat{\uparrow} \dashv \check{\downarrow} \qquad \hat{\uparrow} \dashv \check{\downarrow} \frac{\dot{X} \Vdash \check{\downarrow} \Delta}{\hat{\uparrow} \dot{X} \vdash \Delta}$$

Structural rules

$$\frac{\mathring{X} \overset{\circ}{\vdash} \Delta}{\mathring{X} \overset{\circ}{\vdash} \check{\downarrow} \Delta} \check{\downarrow} \qquad \hat{\uparrow} \frac{X \overset{\circ}{\vdash} \mathring{\Delta}}{\hat{\uparrow} X \overset{\circ}{\vdash} \mathring{\Delta}} \tag{14}$$

▶ **Proposition 14.** *Sequents of the form* $\dot{X} \Vdash Y, \Delta \Vdash \dot{\Gamma}$ *and* $\dot{X} \Vdash \dot{\Delta}$ *are not derivable.*

Proof. By quick induction on the derivation and examination of every rule with the following induction hypothesis: "In a sequent S, if $\dot{X}$ (resp. $\dot{\Delta}$) occurs in S in precedent (resp. succedent) position, then put in display, the succedent (resp. precedent) is either pure negative or shifted positive (resp. pure positive or shifted negative)." It namely works because LG connectives are pure, ℓ, r-variants are shifted and because it holds for the conclusion of rules involving shifts. ◀

Indeed, some conceivable combinations of cut rules in are actually not included in the calculus (see (11)), as well as some conceivable weakening relations are not considered in the algebraic semantics (see definition 7).

3.2 Focalization

In this subsection we first provide a procedural description and a formal definition of *strongly focused proof* of an arbitrary sequent calculus (definition 19, adapted from [21, def. 3]). Then, we show that **fD.LG** has strong focalization (theorem 23). In the end we provide some nomenclature and a diagrammatic representation of the 'topology of rules' of **fD.LG**. We use Ψ, Φ to refer to arbitrary structures.

The backward-looking proof search strategy implemented by a focused sequent calculus (see for instance [2]) can be roughly described as follows: (i) pick a formula, (ii) decompose the chosen formula as much as possible via applications of non-invertible logical rules, (iii) once you reach a subformula of the opposite polarity or an atom, then you may apply structural rules or invertible logical rules, (iv) repeat the process. In order to make precise this informal procedural description, we use a couple of preliminary definitions (see for instance [12]).

▶ **Definition 15** (Signed generation tree). *The positive (resp. negative) generation tree of a structure Ψ, denoted $+\Psi$ (resp. $-\Psi$), is defined by labelling the root node of the generation tree of Ψ with the sign $+$ (resp. $-$), and then propagating the labelling on each remaining node as follows:*

For any node labelled with $h \in \mathcal{F} \cup \mathcal{G}$ of arity $n_h \geq 1$, and for any $1 \leq i \leq n_h$, assign the same (resp. the opposite) sign to its i-th child node if the order-type $\epsilon(h, i) = 1$ (resp. if $\epsilon(h, i) = \partial$).

The signed generation tree of a sequent $\Psi \; t \; \Phi$ consists of the signed generation trees $+\Psi$ and $-\Phi$.

▶ **Definition 16** (Skeleton and PIA). *A node in a signed generation tree of a sequent is called skeleton if it is labelled with $+f$ for some $f \in \mathcal{F}$ or with $-g$ for some $g \in \mathcal{G}$. Otherwise, it is called a PIA node.*

An example of signed generation tree is given in Fig. 3 in appendix C. Notice that any signed generation tree of a well-formed stucture Ψ in the language of **fD.LG** can be partitioned into skeleton vs PIA subtrees (i.e. connected subgraphs of the signed generation tree of Ψ).

▶ **Definition 17** (Transition node). *A transition node of a signed generation tree σ is the uppermost node of a skeleton or PIA subtree excluding the root of σ.*

▶ **Definition 18** (Proof-section). *A proof-section π' of a proof-tree π is a connected subgraph of π, such that for every node $S \in \pi'$, if S is not a leaf of π' and it is introduced by a rule application R, then also the premise(s) of R are in π'.*

▶ **Definition 19** (Strong focalization). *A sequent proof π is* strongly focalized *if cut-free and, for every formula A occurring in π, every PIA subtree of A is constructed by a proof-section of π containing only tonicity rules.*

▶ **Proposition 20.** *Let h be an operational connective occurring in the generation tree of the end-sequent $\Psi \; t \; \Phi$ in a **fD.LG**-proof π, and let S be the uppermost sequent in π where h occurs. If h is a skeleton node, then it is introduced in S via a translation rule. If h is a PIA node, then it is introduced in S via a tonicity rule.*

Proof. Immediate by inspection of the rules of **fD.LG**. ◀

▶ **Proposition 21.** *Let A be a formula occurring in a ℓ-r-variant-free sequent. If a shift labels a node ν of the signed generation tree of A, then either ν is a transition node or it is the root of A.*

Proof. This is due to the presence of shift operators and the polarization of the calculus: For every LG structural connective $\star \in \mathcal{F}$ (resp. $\star \in \mathcal{G}$), (i) the target sort of $\star$ is positive (resp. negative), and (ii) the source sort of the i-th argument of $\star$ is positive (resp. negative) iff $\epsilon(\star, i) = 1$. ◀

The language expansion with ℓ-r-variants guarantees that **fD.LG** enjoys the display property. Indeed, any substructure, no matter if it occurs in a positive, negative or neutral sequent, can be isolated either in precedent or, exclusively, in succedent position. This property is desirable when it comes to prove cut-elimination or develop a general theory for a class of calculi. Nevertheless, allowing structural rules in positive or negative sequents has undesirable consequences on focalization. We argue that confining to ℓ-r-variants-free proofs is harmless in the following sense:

▶ **Proposition 22.** *for every **fD.LG**-derivable sequent S there exists an equivalent ℓ-r-variants-free sequent S' such that S' has a ℓ-r-variants-free proof.*

Proof. We provide here a sketch of the proof. First of all, notice that display postulates are invertible unary structural rules, no other structural rules is allowed in positive or negative sequents, and auxiliary formulas in tonicity rules occur in isolation. Therefore, even tough we may apply a series of display postulates, what we get are equivalent sequents that can be further manipulated only by applications of display postulates. Therefore, the proof search boils down to retrieve back the initial sequent of this list of equivalent sequents and continue as planned. ◀

We can now state the strong focalization property tailored to **fD.LG**.

▶ **Theorem 23.** *Every cut-free and ℓ-r-variants-free proof in **fD.LG** is strongly focalized.*

Proof. Fix a cut-free and ℓ-r-variants-free **fD.LG**-proof π, a formula A occurring a sequent of π, and a PIA subtree Σ of A. We prove by induction on Σ that for every subtree Σ' of Σ which is closed by descendent, the subgraph of π formed by the rules introducing the connectives of Σ' is a proof-section of π of end sequent S, and if $\Sigma' \neq \Sigma$ then S is of the form $(*) : X \vdash P$ or $N \vdash \Delta$.

Call h the root of Σ' and R the rule introducing h in π. We decompose $\Sigma' = h(\Sigma_1, ..., \Sigma_n)$, with $n \in \{1, 2\}$ (h is a shift or LG connective) and Σ_i a subtree closed by descendent. As h is a PIA node, R is a tonicity rule by proposition 20.

Case (a): If Σ_i is empty, we let π_i be the tree consisting of the i-th premise S_i of R. As S_i is derivable, π_i is a proof-section.

Case (b): If Σ_i is non-empty, we apply the induction hypothesis on Σ_i, yielding a proof-section π_i of π containing only tonicity rules and of end sequent S_i of the form $(*)$.

Take π' the subgraph made of π_1,..., π_n and the conclusion S of R. In case (a), S is connected to π_i by construction. In case (b), by looking at the rules, the only variant-free rules applicable on focused sequents (i.e. sequents of the form $(*)$) are tonicity rules, introducing an operational connective. Therefore, the only possibility is that the rule after π_i is R, so S is connected to π_i. Therefore, π' is a proof-section containing only tonicity rules and introducing all connectives of Σ'.

If $\Sigma \neq \Sigma'$, h is not the root of Σ. Therefore, h is not a transition node and not the root of A, so h is not a shift by proposition 21. Therefore, S is also of the form $(*)$.

◀

▶ **Proposition 24.** *Every PIA subtree of a formula occurring in a variant-free **fD.LG**-sequent contains at least one LG-connective.*

Proof. This is due to the fullness of the polarization, i.e. the sort of shifts. The target of $\uparrow$ and $\downarrow$ is shifted but their source sort is pure, i.e. their argument must begin by a LG formula or an atom. In other words, composing $\uparrow$ and $\downarrow$ is impossible. ◀

Proposition 24 forces the focused sections to be uninterrupted from the point of view of LG connectives. In a forward-looking derivation, it is then impossible to defocus a formula A, and then refocus on A. Therefore, translating a **fD.LG** derivation to **f.LG** by removing shift rules would preserve strong focalization.

Now we provide the definition of phases and phase transitions tailored to **fD.LG**.

▶ **Definition 25** (Phases and phase transitions). *Let π be a cut-free and ℓ-r-variant-free proof in **fD.LG**. A sequent S occurring in π is* focused *(aka S is in a focused phase of π) if it is positive ($S = \mathring{X} \mathring{\vdash} \mathring{Y}$) or negative ($S = \mathring{\Delta} \mathring{\vdash} \mathring{\Gamma}$) and no structural shift occurs in S (namely, $\check{\downarrow}, \check{\downarrow}, \hat{\uparrow}, \hat{\uparrow}$). Any other sequent S' occurring in π is* non-focused *(aka S' is in a non-focused phase of π).*

A phase transition *in π is a proof-section π' of π such that the LG-connectives tonicity rules are not applied in π' and its initial-sequent is focused (resp. non-focused) iff its end-sequent is non-focused (resp. focused). A phase transition where the initial-sequent is focused is called* defocusing*, and* focusing *otherwise.*

By design of **fD.LG**, the application of a shift logical rule is needed to move from a focused to a non-focused phase (resp. from a non-focused to a focused phase). Therefore, we may say that principal shifted formulas are the gate-keepers of phase transitions. Because **fD.LG** enjoys subformula property, any formula introduced in a cut-free proof π, and in particular shifted formulas, will also occur in the conclusion of π. Therefore, we may say that shifted formulas are *witnesses* of the relevant proof structure of π. Therefore, we find useful to introduce the following nomenclature:

▶ **Definition 26** (Entry-point and exit-point). *The principal formula introduced by $\downarrow_L$ (resp. $\uparrow_R$) is called the positive (resp. negative)* entry-point *of the induced phase transition. The principal formula introduced by $\downarrow_R$ (resp. $\uparrow_L$) is called the positive (resp. negative)* exit-point *of the induced phase transition.*

We now provide a diagrammatic representation to visualise the topology of rules and phase transitions tailored to **fD.LG** and, ultimately, facilitate the proof of strong focalization.

The diagram in figure 2 depicts the phase transition flow chart of **fD.LG**. The white area contains the generic form of focused sequents, where each of them is either positive or negative (see definition 25). The grey and yellow areas contain the generic form of non-focused sequents (see definition 25), where all neutral sequent seat inside the yellow area.[2] We use metavariables S, S', S'' for sequents and arrows to depict rules. Arrows $\overset{R}{\rightarrow}$ and $\circ\overset{R}{\rightarrow}$ points towards the conclusion of the rule R. In particular, $\overset{R}{\rightarrow} S$ represents a zeroary rule R (i.e. an axiom) and $S' \overset{R}{\rightarrow} S$ represents a unary rule R (here shift logical rules). A double-headed arrow $\overset{R}{\leftrightarrow}$ represents an invertible rule (here structural rules introducing or eliminating a shift). $\multimap$ are 'teleporters' where the configuration $S' \multimap$ and $S'' \multimap$ together with $\circ\!\!\rightarrow S$ represents a binary rule with premises S', S'' and conclusion S (i.e. tonicity rules).[3] To exemplify the conventions involving teleporters, let us consider two configurations included in the diagram of figure 2. (i) Sequents of the form $X \vdash P$ could occur as premises and conclusion of $\otimes_R$, therefore they occurs in the configuration $X \vdash P \multimap$ and $X \vdash P \multimap$ together with $\circ\!\!\rightarrow X \vdash P$. (ii) Sequents of the form $X \vdash P$ could occur as premise of $\backslash_L$ and sequents of the form $N \vdash \Delta$ could occur as premise and conclusion of $\backslash_L$, therefore they occur in the configuration $X \vdash P \multimap$ and $N \vdash \Delta \multimap$ together with $\circ\!\!\rightarrow N \vdash \Delta$.

Summing up, the topology of rules is a follows: (i) the white area is closed under axioms, tonicity rules, and display postulates for ℓ-r-variants, (ii) the gray area is closed under display postulates for shifts and ℓ-r-variants, (iii) the yellow area is closed under any other structural rules (i.e. display postulates for LG-connectives and, whenever we consider analytic extensions of the minimal logic, all the relevant additional structural rules) and translation rules, (iv) the boundary between white and gray areas is crossed only by (non-invertible) shift logical rules, and (v) the boundary between gray and yellow area is crossed only by (invertible) shift structural rules.

---

[2] Notice that sequents of the form $\dot{X} \dot{\vdash} Y, \Delta \dot{\vdash} \dot{\Gamma}$ and $\dot{X} \dot{\vdash} \dot{\Delta}$ are not derivable (see proposition 14) and, therefore, they are not included in the diagram.

[3] Notice that in this case we do not explicitly mention the name of the rule in the diagram.

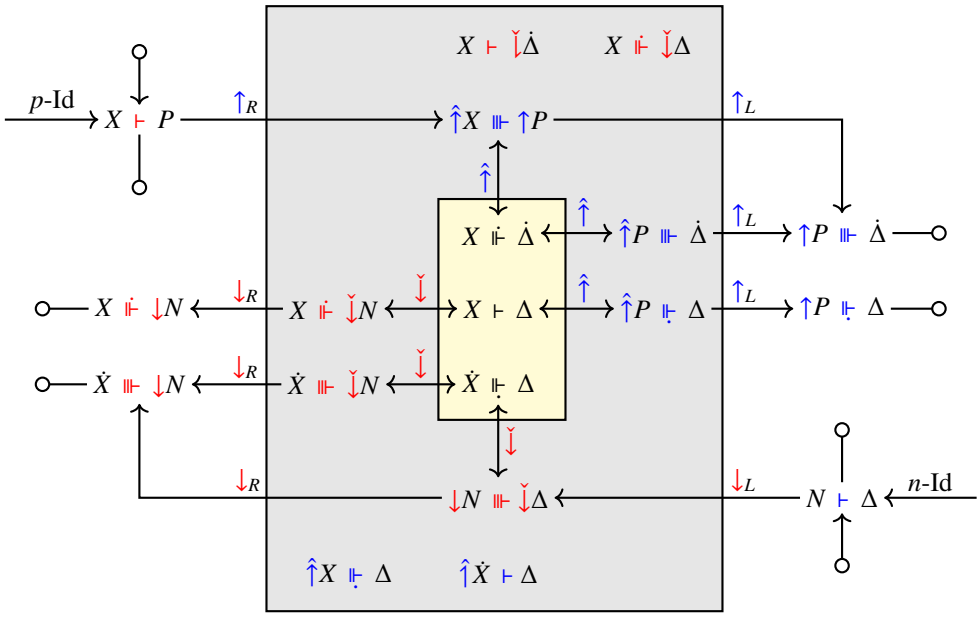


Figure 2 The topology of **fD.LG**-rules and phase transitions.

4 Soundness and completeness

Soundness and completeness are proven as usual in the case of algebraic semantics, where the only departure is that now we consider weakening relations instead of just orders. Soundness is stated as follows:

▶ **Theorem 27.** *Each rule of the focused display calculus **fD.LG** is sound under any interpretation in a fully polarized algebras* $\mathbb{FP}.\mathbb{LG}$.

Proof. Given a $\mathbb{FP}.\mathbb{LG}$-algebra $\mathbb{A}$ and an interpretation $(\cdot)^{\mathbb{A}}$, it is straightforward to check by induction on the complexity of proofs that for every sequent S derivable in **fD.LG**, the interpretation $(S)^{\mathbb{A}}$ is valid. We leave the proof to the reader. Below we simply recall that interpretations of pure atomic formulas $p^{\mathbb{A}}$ and $n^{\mathbb{A}}$ homomorphically extend to arbitrary formulas, and each consequence relation is interpreted by a weakening relation as follows

t	$\vdash$	$\dot{\vdash}$	$\Vdash$	$\vdash$	$\dot{\vdash}$	$\Vdash$	$\vdash$	$\dot{\vdash}$	$\dot{\Vdash}$	$\mathring{\vdash}$	$\mathring{\vdash}$	$\mathring{\vdash}$
$t^{\mathbb{A}}$	$\leq$	$\geq$	$\leqq$	$\leq$	$\geq$	$\leqq$	$\leq$	$\geq$	$\geqq$	$\leq^{\circ}$	$\geq^{\circ}$	$\leq^{\circ}$

(15)

◀

In order to prove completeness, we need to introduce the auxiliary notion of standard sequents.

▶ **Definition 28.** *The principal subtree of a structure* Ψ *is the largest subtree of the signed generation tree of A containing the root and which is either a skeleton subtree or a PIA subtree.*

▶ **Definition 29.** *Let* Ψ *be a structure. We call* $\lfloor\!\lfloor \Psi \rfloor\!\rfloor$ *(resp.* $\lceil\!\lceil \Psi \rceil\!\rceil$*) the structure of same sort obtained, when it is defined, by turning every connective of its principal subtree* Σ *into either*

(i) *its structural counterpart if* Σ *is a skeleton subtree of* $+\Psi$ *(resp.* $-\Psi$*)*

(ii) *its operational counterpart if* Σ *is a PIA subtree of* $+\Psi$ *(resp.* $-\Psi$*)*

and turning all other connectives into their operational counterpart.

Given a well-formed sequent $\Psi\ t\ \Phi$*, its* ***standard sequent*** *is* $\lfloor\!\lfloor \Psi \rfloor\!\rfloor\ t\ \lceil\!\lceil \Phi \rceil\!\rceil$.

To exemplify the instrumental use of standard sequents in proving completeness, consider the following observation. The sequent $p \otimes q \vdash p \otimes q$ is not derivable in **fD.LG**, despite the fact that $\leq$ is a partial order, so in particular $p^{\mathbb{P}} \otimes^{\mathbb{P}} q^{\mathbb{P}} \leq p^{\mathbb{P}} \otimes^{\mathbb{P}} q^{\mathbb{P}}$ in every $\mathbb{FP}.\mathbb{LG}$-algebra. However, the standard sequent $p \hat{\otimes} q \vdash p \otimes q$ is **fD.LG**-derivable and moreover $(p \hat{\otimes} q)^{\mathbb{P}} = (p \otimes q)^{\mathbb{P}} = p^{\mathbb{P}} \otimes^{\mathbb{P}} q^{\mathbb{P}}$. See lemma 31 in appendix A for a recursive definition of $\lfloor\!\lfloor\ \rfloor\!\rfloor$ and $\lceil\!\lceil\ \rceil\!\rceil$. Completeness is stated as follows:

▶ **Theorem 30.** *For every* $\mathbb{FP}.\mathbb{LG}$*-algebra* $\mathbb{A}$ *and every well-formed sequent* $\Psi\ t\ \Phi$ *in the language of* ***fD.LG****, if the interpretation* $(\Psi\ t\ \Phi)^{\mathbb{A}}$ *is a valid, then the standard sequent* $\lfloor\!\lfloor \Psi \rfloor\!\rfloor\ t\ \lceil\!\lceil \Phi \rceil\!\rceil$ *is derivable in* ***fD.LG****.*

Proof. We prove completeness by building a syntactic model $\mathbb{A}$. Let $\approx_{\mathbb{P}}$, $\approx_{\dot{\mathbb{P}}}$, $\approx_{\mathbb{N}}$ and $\approx_{\dot{\mathbb{N}}}$ be the equivalence relation generated by (16).

$$\begin{array}{llllll} \Psi \approx_{\mathbb{P}} \Phi & \text{iff} & \lfloor\!\lfloor \Psi \rfloor\!\rfloor \vdash \lceil\!\lceil \Phi \rceil\!\rceil & \text{and} & \lfloor\!\lfloor \Phi \rfloor\!\rfloor \vdash \lceil\!\lceil \Psi \rceil\!\rceil \\ \Psi \approx_{\dot{\mathbb{P}}} \Phi & \text{iff} & \lfloor\!\lfloor \Psi \rfloor\!\rfloor \Rrightarrow \lceil\!\lceil \Phi \rceil\!\rceil & \text{and} & \lfloor\!\lfloor \Phi \rfloor\!\rfloor \Rrightarrow \lceil\!\lceil \Psi \rceil\!\rceil \\ \Psi \approx_{\mathbb{N}} \Phi & \text{iff} & \lfloor\!\lfloor \Psi \rfloor\!\rfloor \vdash \lceil\!\lceil \Phi \rceil\!\rceil & \text{and} & \lfloor\!\lfloor \Phi \rfloor\!\rfloor \vdash \lceil\!\lceil \Psi \rceil\!\rceil \\ \Psi \approx_{\dot{\mathbb{N}}} \Phi & \text{iff} & \lfloor\!\lfloor \Psi \rfloor\!\rfloor \Rrightarrow \lceil\!\lceil \Phi \rceil\!\rceil & \text{and} & \lfloor\!\lfloor \Phi \rfloor\!\rfloor \Rrightarrow \lceil\!\lceil \Psi \rceil\!\rceil \end{array} \tag{16}$$

In particular, $\approx_{\mathbb{P}}$, $\approx_{\dot{\mathbb{P}}}$, $\approx_{\mathbb{N}}$ and $\approx_{\dot{\mathbb{N}}}$ are congruence relations (by tonicity rules and cut rules, see appendix A for a detailed proof). For any $s \in \{\mathbb{P}, \dot{\mathbb{P}}, \mathbb{N}, \dot{\mathbb{N}}\}$, let $[\Psi]_{\approx_s}$ denote the class of structures Φ such that $\Phi \approx_s \Psi$. We define operations and weakening relations by (17).

$$\begin{array}{c} [\mathring{X}]_{\approx_{\dot{\mathbb{P}}}} \otimes^{\mathbb{A}} [\mathring{Y}]_{\approx_{\dot{\mathbb{P}}}} = [\mathring{X} \hat{\otimes} \mathring{Y}]_{\approx_{\mathbb{P}}} \quad [\mathring{X}]_{\approx_{\dot{\mathbb{P}}}} \oslash^{\mathbb{A}} [\mathring{\Delta}]_{\approx_{\dot{\mathbb{N}}}} = [\mathring{X} \hat{\oslash} \mathring{\Delta}]_{\approx_{\mathbb{P}}} \quad [\mathring{\Delta}]_{\approx_{\dot{\mathbb{N}}}} \otimes\!\!\!\!\diagup^{\mathbb{A}} [\mathring{\Delta}]_{\approx_{\dot{\mathbb{P}}}} = [\mathring{\Delta} \hat{\otimes\!\!\!\!\diagup} \mathring{Y}]_{\approx_{\mathbb{P}}} \\ [\mathring{\Delta}]_{\approx_{\dot{\mathbb{N}}}} \oplus^{\mathbb{A}} [\mathring{\Gamma}]_{\approx_{\dot{\mathbb{N}}}} = [\mathring{\Delta} \check{\oplus} \mathring{\Gamma}]_{\approx_{\mathbb{N}}} \quad [\mathring{X}]_{\approx_{\dot{\mathbb{P}}}} \setminus^{\mathbb{A}} [\mathring{\Delta}]_{\approx_{\dot{\mathbb{N}}}} = [\mathring{X} \check{\setminus} \mathring{\Delta}]_{\approx_{\mathbb{N}}} \quad [\mathring{\Delta}]_{\approx_{\dot{\mathbb{N}}}} /^{\mathbb{A}} [\mathring{\Delta}]_{\approx_{\dot{\mathbb{P}}}} = [\mathring{\Delta} \check{/} \mathring{Y}]_{\approx_{\mathbb{N}}} \\ \downarrow^{\mathbb{A}} [\Delta]_{\approx_{\mathbb{N}}} = [\check{\downarrow} \Delta]_{\approx_{\mathbb{P}}} \quad \downarrow^{\mathbb{A}} [\dot{\Delta}]_{\approx_{\dot{\mathbb{N}}}} = [\check{\downarrow} \dot{\Delta}]_{\approx_{\dot{\mathbb{P}}}} \quad \uparrow^{\mathbb{A}} [X]_{\approx_{\mathbb{P}}} = [\hat{\uparrow} X]_{\approx_{\mathbb{N}}} \quad \uparrow^{\mathbb{A}} [X]_{\approx_{\dot{\mathbb{P}}}} = [\hat{\uparrow} X]_{\approx_{\mathbb{N}}} \\ \text{and similar for the } \ell, r\text{-variants} \\ \text{or a turnstile } t \text{ from sort } s \text{ to } s', [\Psi]_{\approx_s}\ t^{\mathbb{A}}\ [\Phi]_{\approx_{s'}} \text{iff } \lfloor\!\lfloor \Psi \rfloor\!\rfloor\ t\ \lceil\!\lceil \Phi \rceil\!\rceil \text{ is derivable} \end{array} \tag{17}$$

It is not difficult to see that the operations and relations of (17) are well-defined, and that the relations are indeed orders or weakening relations. The technical proof is provided in appendix A.

We take $\mathbb{P} = \mathsf{PurePosStr}/\approx_{\mathbb{P}}$, $\dot{\mathbb{P}} = \mathsf{ShiftedPosStr}/\approx_{\dot{\mathbb{P}}}$, $\mathbb{N} = \mathsf{PureNegStr}/\approx_{\mathbb{N}}$ and $\dot{\mathbb{N}} = \mathsf{ShiftedNegStr}/\approx_{\dot{\mathbb{N}}}$. It is easy to show that properties (4), (5), (6) and (7) of definition 7 are verified thanks to the correspondent rules of the calculus. ◀

5 Conclusions

We observe that every connective in the language of **fD.LG** exhibits a core of minimal properties in any sub-algebra, namely it has finite arity and it is residuated in each coordinate. Nevertheless, we do not assume that a composition of shifts gives an identity. This leaves open the option that additional properties hold in the full algebra. Special sub-classes of $\mathbb{FP}.\mathbb{LG}$ algebras could then be captured by expanding the minimal calculus with opportune structural rules. If the needed rules are analytic-inductive and involve only LG-connectives, we conjecture that the cut-elimination will be preserved too. What we have in mind here is a natural generalisation of the cut-elimination meta-theorem of multi-type display calculi for a broader class of calculi, of which **fD.LG** is a prototypical example. We also plan to investigate up to which extent focalization will be preserved too.

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A Proof complements

Proof of proposition 4. Set $A_1, A_2 \in \mathcal{A}$, and $B'_1, B'_2 \in \mathcal{B}'$ such that $A_2 \leq_{\mathcal{A}} A_1$, $B'_1 \leq_{\mathcal{B}'} B'_2$ and $A_1 \preccurlyeq B_1$. Equation $A_1 \preccurlyeq B_1$ is equivalent to $L(A_1) \preccurlyeq_{\mathcal{B}} B'_1$, and as $\preccurlyeq_{\mathcal{B}}$ is a weakening relation, we have $L(A_1) \preccurlyeq_{\mathcal{B}} B'_2$. This last equation is equivalent (by adjunction) to $A_1 \preccurlyeq_{\mathcal{A}} R(B'_2)$ and as $\preccurlyeq_{\mathcal{A}}$ is a weakening relation, we have $A_2 \preccurlyeq_{\mathcal{A}} R(B'_2)$, which is equivalent to $A_2 \preccurlyeq B'_2$. Therefore, $\preccurlyeq$ is a weakening relation. ◀

Proof that $\mathring{\leq}$ is a weakening relation (definition 6). We have $\mathring{\leq} = \leqq \sqcup \leq \sqcup \overline{\overline{\leq}}$ a relation on $(\mathcal{A} \sqcup \mathcal{A}') \times (\mathcal{B} \sqcup \mathcal{B}')$. Set $A_1, A_2 \in \mathcal{A} \sqcup \mathcal{A}'$ and $B_1, B_2 \in \mathcal{B} \sqcup \mathcal{B}'$ such that $A_2 \leq_{\mathcal{A}\sqcup\mathcal{A}'} A_1$, $B_1 \leq_{\mathcal{B}\sqcup\mathcal{B}'} B_2$ and $A_1 \mathring{\leq} B_1$. We only show how to get $A_2 \mathring{\leq} B_1$, the other side is similar by symmetry of the problem.

- If A_1 and A_2 are both in $\mathcal{A}$ (resp. $\mathcal{A}'$), then $A_2 \leq_{\mathcal{A}\sqcup\mathcal{A}'} A_1$ is equivalent to $A_2 \leq_{\mathcal{A}} A_1$ (resp. $A_2 \leq_{\mathcal{A}'} A_1$) and $A_1 \mathring{\leq} B_1$ is equivalent to $A_1 \overline{\overline{\leq}} B_1$ or $A_1 \leq B_1$ (resp. $A_1 \leqq B_1$). In all cases, because $\leqq, \leq$ and $\overline{\overline{\leq}}$ are all weakening relations, we have $A_2 \leqq B_1$, $A_2 \leq B_1$ or $A_2 \overline{\overline{\leq}} B_1$, hence $A_2 \mathring{\leq} B_1$.
- If $A_2 \in \mathcal{A}$ and $A_1 \in \mathcal{A}'$, then we have $A_2 \preccurlyeq_{\mathcal{A}} A_1$ and $A_1 \leqq B_1$. Therefore $A_2 \leq B_1$, because $\preccurlyeq_{\mathcal{A}} \leqq \subseteq \leq$ by hypothesis, hence $A_2 \mathring{\leq} B_1$.
- The case $A_2 \in \mathcal{A}'$ and $A_1 \in \mathcal{A}$ is impossible because the union is disjoint and $\preccurlyeq_{\mathcal{A}}$ is only from $\mathcal{A}$ to $\mathcal{A}'$.

◀

The following four lemmas provide detailed complements to the proof of completeness by showing that the equivalence relations $\approx_s$ for $s \in \{\mathbb{P}, \dot{\mathbb{P}}, \mathbb{N}, \dot{\mathbb{N}}\}$ are congruences are thus that the operations and weakening relations / orders defined on them are well-defined.

▶ **Lemma 31.** *Given a formula A and a structure Ψ, we write* $\mathsf{Str}(A)$ *the structure obtained by turning the connectives of A into their structural counterparts, and* $\mathsf{Fm}(\Psi)$ *the formula obtained by turning the connectives of Ψ into their operational counterpart, when it exists (in the other case,*

$\mathsf{Fm}(\Psi)$ *is not defined). When* $\llbracket\ \rrbracket$ *and* $\lceil\!\lceil\ \rceil\!\rceil$ *are defined, they enjoy the following property:*

$$\begin{array}{c}
\llbracket A\rrbracket = \llbracket \mathsf{Str}(A)\rrbracket \quad \textit{if } A \textit{ is a } \mathcal{F}\textit{-formula} \\
\llbracket \Psi\rrbracket = \mathsf{Fm}(\Psi) \quad \textit{if } \Psi \textit{ is a } \mathcal{G}\textit{-structure} \\
\llbracket \mathring{X} \mathbin{\hat{\otimes}} \mathring{Y}\rrbracket = \llbracket \mathring{X}\rrbracket \mathbin{\hat{\otimes}} \llbracket \mathring{Y}\rrbracket \qquad \llbracket \mathring{X} \mathbin{\hat{\oslash}} \mathring{\Delta}\rrbracket = \llbracket \mathring{X}\rrbracket \mathbin{\hat{\oslash}} \lceil\!\lceil \mathring{\Delta}\rceil\!\rceil \qquad \llbracket \mathring{\Delta} \mathbin{\hat{\otimes}} \mathring{Y}\rrbracket = \lceil\!\lceil \mathring{\Delta}\rceil\!\rceil \mathbin{\hat{\otimes}} \llbracket \mathring{Y}\rrbracket \\
\llbracket \mathring{\Gamma} \mathbin{\hat{\otimes}}_\ell \mathring{Y}\rrbracket = \llbracket \mathring{\Gamma}\rrbracket \mathbin{\hat{\otimes}}_\ell \llbracket \mathring{Y}\rrbracket \qquad \llbracket \mathring{\Gamma} \mathbin{\hat{\oslash}}_\ell \mathring{\Delta}\rrbracket = \llbracket \mathring{\Gamma}\rrbracket \mathbin{\hat{\oslash}}_\ell \lceil\!\lceil \mathring{\Delta}\rceil\!\rceil \qquad \llbracket \mathring{X} \mathbin{\hat{\otimes}}_\ell \mathring{Y}\rrbracket = \lceil\!\lceil \mathring{X}\rceil\!\rceil \mathbin{\hat{\otimes}}_\ell \llbracket \mathring{Y}\rrbracket \\
\llbracket \mathring{X} \mathbin{\hat{\otimes}}_r \mathring{\Gamma}\rrbracket = \llbracket \mathring{X}\rrbracket \mathbin{\hat{\otimes}}_r \llbracket \mathring{\Gamma}\rrbracket \qquad \llbracket \mathring{X} \mathbin{\hat{\oslash}}_r \mathring{Y}\rrbracket = \llbracket \mathring{X}\rrbracket \mathbin{\hat{\oslash}}_r \lceil\!\lceil \mathring{Y}\rceil\!\rceil \qquad \llbracket \mathring{\Delta} \mathbin{\hat{\otimes}}_r \mathring{\Gamma}\rrbracket = \lceil\!\lceil \mathring{\Delta}\rceil\!\rceil \mathbin{\hat{\otimes}}_r \llbracket \mathring{\Gamma}\rrbracket \\
\llbracket \hat{\uparrow} X\rrbracket = \hat{\uparrow} \llbracket X\rrbracket \qquad\qquad \llbracket \hat{\uparrow} \dot{X}\rrbracket = \hat{\uparrow} \llbracket \dot{X}\rrbracket \\
\lceil\!\lceil A\rceil\!\rceil = \lceil\!\lceil \mathsf{Str}(A)\rceil\!\rceil \quad \textit{if } A \textit{ is a } \mathcal{G}\textit{-formula} \\
\lceil\!\lceil \Psi\rceil\!\rceil = \mathsf{Fm}(\Psi) \quad \textit{if } \Psi \textit{ is a } \mathcal{F}\textit{-structure} \\
\lceil\!\lceil \mathring{\Delta} \mathbin{\check{\oplus}} \mathring{\Gamma}\rceil\!\rceil = \lceil\!\lceil \mathring{\Delta}\rceil\!\rceil \mathbin{\check{\oplus}} \lceil\!\lceil \mathring{\Gamma}\rceil\!\rceil \qquad \lceil\!\lceil \mathring{X} \mathbin{\check{\setminus}} \mathring{\Delta}\rceil\!\rceil = \llbracket \mathring{X}\rrbracket \mathbin{\check{\setminus}} \lceil\!\lceil \mathring{\Delta}\rceil\!\rceil \qquad \lceil\!\lceil \mathring{\Delta} \mathbin{\check{/}} \mathring{Y}\rceil\!\rceil = \lceil\!\lceil \mathring{\Delta}\rceil\!\rceil \mathbin{\check{/}} \llbracket \mathring{Y}\rrbracket \\
\lceil\!\lceil \mathring{Y} \mathbin{\check{\oplus}}_\ell \mathring{\Gamma}\rceil\!\rceil = \lceil\!\lceil \mathring{Y}\rceil\!\rceil \mathbin{\check{\oplus}}_\ell \lceil\!\lceil \mathring{\Gamma}\rceil\!\rceil \qquad \lceil\!\lceil \mathring{\Gamma} \mathbin{\check{\setminus}}_\ell \mathring{\Delta}\rceil\!\rceil = \llbracket \mathring{\Gamma}\rrbracket \mathbin{\check{\setminus}}_\ell \lceil\!\lceil \mathring{\Delta}\rceil\!\rceil \qquad \lceil\!\lceil \mathring{X} \mathbin{\check{/}}_\ell \mathring{Y}\rceil\!\rceil = \lceil\!\lceil \mathring{X}\rceil\!\rceil \mathbin{\check{/}}_\ell \llbracket \mathring{Y}\rrbracket \\
\lceil\!\lceil \mathring{\Delta} \mathbin{\check{\oplus}}_r \mathring{Y}\rceil\!\rceil = \lceil\!\lceil \mathring{\Delta}\rceil\!\rceil \mathbin{\check{\oplus}}_r \lceil\!\lceil \mathring{Y}\rceil\!\rceil \qquad \lceil\!\lceil \mathring{X} \mathbin{\check{\setminus}}_r \mathring{Y}\rceil\!\rceil = \llbracket \mathring{X}\rrbracket \mathbin{\check{\setminus}}_r \lceil\!\lceil \mathring{Y}\rceil\!\rceil \qquad \lceil\!\lceil \mathring{\Delta} \mathbin{\check{/}}_r \mathring{\Gamma}\rceil\!\rceil = \lceil\!\lceil \mathring{\Delta}\rceil\!\rceil \mathbin{\check{/}}_r \llbracket \mathring{\Gamma}\rrbracket \\
\lceil\!\lceil \check{\downarrow} \Delta\rceil\!\rceil = \check{\downarrow} \lceil\!\lceil \Delta\rceil\!\rceil \qquad\qquad \lceil\!\lceil \check{\downarrow} \dot{\Delta}\rceil\!\rceil = \check{\downarrow} \lceil\!\lceil \dot{\Delta}\rceil\!\rceil \\
\llbracket p\rrbracket = \lceil\!\lceil p\rceil\!\rceil = p \qquad \llbracket n\rrbracket = \lceil\!\lceil n\rceil\!\rceil = n
\end{array}$$

Proof. Unfolding definition 29 directly gives these results. ◂

▸ **Lemma 32.** *If* $\mathring{X} \vdash \Delta$ *(resp.* $X \vdash \mathring{\Delta}$*) is derivable, then* $\dfrac{\mathring{X} \vdash \Delta}{\mathring{X} \vdash \mathsf{Fm}(\Delta)}$ $\left(\textit{resp.}\ \dfrac{X \vdash \mathring{\Delta}}{\mathsf{Fm}(X) \vdash \mathring{\Delta}}\right)$ *is derivable.*

Proof. By induction on Δ (resp. X), by successively applying translation rules on all structural connectives of Δ (resp. X). ◂

▸ **Lemma 33.** *For every structure* Ψ *of sort* s*, the sequent* $\llbracket\Psi\rrbracket\ t_s\ \lceil\!\lceil\Psi\rceil\!\rceil$ *is derivable, if it is defined.*

Proof. By induction on Ψ. If Ψ is an atomic formula, $\llbracket\Psi\rrbracket = \lceil\!\lceil\Psi\rceil\!\rceil$ so p-Id or n-Id is applicable.

If $\Psi = \mathring{X} \mathbin{\hat{\otimes}} \mathring{Y}$, let us develop the case $\mathring{X} = X$ and $\mathring{Y} = \dot{Y}$, the others being similar. We use the induction hypothesis on X and $\dot{Y}$. By assumption, $\llbracket\dot{Y}\rrbracket$ exists, so $\dot{Y}$ does begin by $\check{\oplus}_\ell$, $\check{\oplus}_r$, $\check{\setminus}_\ell$, $\check{\setminus}_r$, $\check{/}_\ell$ or $\check{/}_r$. So it begins by a shift: $\dot{Y} = \check{\downarrow}\Delta$ and we can then derive

$$\dfrac{\dfrac{(\mathrm{IH})_X}{\llbracket X\rrbracket \vdash \lceil\!\lceil X\rceil\!\rceil}\ \text{Lemma 31}}{\llbracket X\rrbracket \vdash \mathsf{Fm}(X)} \qquad \dfrac{\dfrac{\dfrac{\dfrac{\dfrac{\dfrac{(\mathrm{IH})_{\dot{Y}}}{\llbracket \dot{Y}\rrbracket \Rrightarrow \lceil\!\lceil \check{\downarrow}\Delta\rceil\!\rceil}\ \text{Lemma 31}}{\llbracket \dot{Y}\rrbracket \Rrightarrow \check{\downarrow}\lceil\!\lceil \Delta\rceil\!\rceil}\ \check{\downarrow}}{\llbracket \dot{Y}\rrbracket \Vdash \lceil\!\lceil \Delta\rceil\!\rceil}\ \text{Lemma 32}}{\llbracket \dot{Y}\rrbracket \Vdash \mathsf{Fm}(\Delta)}\ \check{\downarrow}}{\llbracket \dot{Y}\rrbracket \Rrightarrow \check{\downarrow}\mathsf{Fm}(\Delta)}\ \downarrow_R}{\llbracket \dot{Y}\rrbracket \Rrightarrow \downarrow\mathsf{Fm}(\Delta)} \qquad \llbracket \dot{Y}\rrbracket \Rrightarrow \mathsf{Fm}(\check{\downarrow}\Delta)$$

$$\dfrac{\llbracket X\rrbracket \vdash \mathsf{Fm}(X) \qquad \llbracket \dot{Y}\rrbracket \Rrightarrow \downarrow\mathsf{Fm}(\Delta) \quad \llbracket \dot{Y}\rrbracket \Rrightarrow \mathsf{Fm}(\check{\downarrow}\Delta)}{\llbracket X\rrbracket \mathbin{\hat{\otimes}} \llbracket \dot{Y}\rrbracket \vdash \mathsf{Fm}(X) \otimes \mathsf{Fm}(\check{\downarrow}\Delta)}\ \otimes_R}{\llbracket X \mathbin{\hat{\otimes}} \dot{Y}\rrbracket \vdash \lceil\!\lceil X \mathbin{\hat{\otimes}} \check{\downarrow}\Delta\rceil\!\rceil}\ \text{Lemma 31}$$

The other LG-connectives work similarly.

If $\Psi = \check{\downarrow}\Delta$, we use the induction hypothesis on Δ to get

$$\dfrac{\dfrac{\dfrac{\llbracket \Delta\rrbracket \vdash \lceil\!\lceil \Delta\rceil\!\rceil}{\mathsf{Fm}(\Delta) \vdash \lceil\!\lceil \Delta\rceil\!\rceil}\ \text{Lemma 31}}{\downarrow\mathsf{Fm}(\Delta) \Rrightarrow \check{\downarrow}\lceil\!\lceil \Delta\rceil\!\rceil}\ \downarrow_L}{\llbracket \check{\downarrow}\Delta\rrbracket \Rrightarrow \lceil\!\lceil \check{\downarrow}\Delta\rceil\!\rceil}\ \text{Lemma 31}$$

$\hat{\uparrow}$ works dually.

If Ψ begins by a ℓ, r-variants or shift adjoint, Ψ is not in the domain of both $\llbracket\ \rrbracket$ and $\lceil\!\lceil\ \rceil\!\rceil$. ◀

▶ **Lemma 34.** *For every derivable sequents $\llbracket\Psi\rrbracket\ t\ \lceil\!\lceil\Phi\rceil\!\rceil$ and $\llbracket\Phi\rrbracket\ t'\ \lceil\!\lceil\Psi'\rceil\!\rceil$, we can derive the cut*

$$\frac{\llbracket\Psi\rrbracket\ t\ \lceil\!\lceil\Phi\rceil\!\rceil \qquad \llbracket\Phi\rrbracket\ t'\ \lceil\!\lceil\Psi'\rceil\!\rceil}{\llbracket\Psi\rrbracket\ tt'\ \lceil\!\lceil\Psi'\rceil\!\rceil}$$

where the composition tt' is determined by the sort of Ψ and Ψ'.

Proof. We proceed by induction on Φ.

If Φ is an atomic formula, $\llbracket\Phi\rrbracket = \lceil\!\lceil\Phi\rceil\!\rceil$ is a formula, so we can proceed to a tt' cut of (11) (i.e. P-Cut, N-Cut, Pn-Cut or nN-Cut).

If $\Phi = \mathring{X} \hat{\otimes} \mathring{Y}$, we know that at the introduction of $\lceil\!\lceil\Psi\rceil\!\rceil = \mathsf{Fm}(\mathring{X}) \otimes \mathsf{Fm}(\mathring{Y})$, we have some sequent $\mathring{X}' \hat{\otimes} \mathring{Y}' \vdash \mathsf{Fm}(\mathring{X}) \otimes \mathsf{Fm}(\mathring{Y})$ and the proof

$$\dfrac{\dfrac{\vdots\,\pi_1 \qquad \vdots\,\pi_2}{\mathring{X}' \mathring{\vdash} \lceil\!\lceil\mathring{X}\rceil\!\rceil \qquad \mathring{Y}' \mathring{\vdash} \lceil\!\lceil\mathring{Y}\rceil\!\rceil}\otimes_R}{\vdots\,\pi}{\mathring{X}' \hat{\otimes} \mathring{Y}' \vdash \lceil\!\lceil\mathring{X}\rceil\!\rceil \otimes \lceil\!\lceil\mathring{Y}\rceil\!\rceil}$$
$$\llbracket\Psi\rrbracket\ t\ \lceil\!\lceil\Phi\rceil\!\rceil$$

We can then apply the induction hypothesis on $\mathring{X}$ and $\mathring{Y}$. Here, we develop case where $t' = \mathring{\vdash}$ (so Ψ is positive):

$$\dfrac{\dfrac{\vdots\,\pi_1 \qquad \dfrac{\dfrac{\dfrac{\vdots\,\pi_2 \qquad \dfrac{\dfrac{\vdots}{\llbracket\mathring{X}\hat{\otimes}\mathring{Y}\rrbracket \mathring{\vdash} \lceil\!\lceil\Psi'\rceil\!\rceil}}{\llbracket\mathring{X}\rrbracket \hat{\otimes} \llbracket\mathring{Y}\rrbracket \mathring{\vdash} \lceil\!\lceil\Psi'\rceil\!\rceil}\ \text{Lemma 31}}{\mathring{Y}' \mathring{\vdash} \lceil\!\lceil\mathring{Y}\rceil\!\rceil \qquad \llbracket\mathring{Y}\rrbracket \mathring{\vdash} \llbracket\mathring{X}\rrbracket \mathbin{\check{\backslash}}_r \lceil\!\lceil\Psi'\rceil\!\rceil}\ \hat{\otimes} \dashv \mathbin{\check{\backslash}}_r}{\mathring{Y}' \mathring{\vdash} \llbracket\mathring{X}\rrbracket \mathbin{\check{\backslash}}_r \lceil\!\lceil\Psi'\rceil\!\rceil}\ (\mathrm{IH})_{\mathring{Y}}}{\llbracket\mathring{X}\rrbracket \hat{\otimes} \llbracket\mathring{Y}\rrbracket \mathring{\vdash} \lceil\!\lceil\Psi'\rceil\!\rceil}\ \hat{\otimes} \dashv \mathbin{\check{/}}_\ell}{\mathring{X}' \mathring{\vdash} \lceil\!\lceil\mathring{X}\rceil\!\rceil \qquad \llbracket\mathring{X}\rrbracket \mathring{\vdash} \lceil\!\lceil\Psi'\rceil\!\rceil \mathbin{\check{/}}_\ell \mathring{Y}'}\ (\mathrm{IH})_{\mathring{X}}}{\mathring{X}' \mathring{\vdash} \lceil\!\lceil\Psi'\rceil\!\rceil \mathbin{\check{/}}_\ell \mathring{Y}'}\ \hat{\otimes} \dashv \mathbin{\check{/}}_\ell}{\mathring{X}' \hat{\otimes} \mathring{Y}' \mathring{\vdash} \lceil\!\lceil\Psi'\rceil\!\rceil}$$
$$\llbracket\Psi\rrbracket\ tt'\ \lceil\!\lceil\Psi'\rceil\!\rceil$$

The case where $t' = \mathring{\vdash}$ works similarly with $\mathbin{\check{\backslash}}$ and $\mathbin{\check{/}}$.

The fact that the uniform substitution $\pi[\lceil\!\lceil\Psi'\rceil\!\rceil/\lceil\!\lceil\Phi\rceil\!\rceil]$ (were $\lceil\!\lceil\Psi'\rceil\!\rceil$ may have a different sort from $\lceil\!\lceil\Phi\rceil\!\rceil$) can be defined and is derivable in **fD.LG** is not proven here. We would have to provide a way of transforming some structural connectives (in specific positions) into others, what is partly already implicit in the use of overloaded connectives (e.g. $\hat{\otimes}$'s arguments can be either pure or shifted). This operation pertains to the problem of canonical cut-elimination with heterogeneous sequents, what is left to a subsequent paper.

If $\Phi = \hat{\uparrow}X$, we use the same procedure, by induction hypothesis on X. The turnstile t' can only be $\mathring{\vdash}$. Here we develop the case where Ψ' is shifted.

$$\dfrac{\begin{array}{c}\vdots\,\pi'\\ X' \vdash \llbracket X\rrbracket\end{array}}{\hat{\uparrow}X' \Vdash \uparrow\llbracket X\rrbracket}\otimes_R \quad\rightsquigarrow\quad \dfrac{\begin{array}{c}\vdots\,\pi'\\ X' \vdash \llbracket X\rrbracket\end{array} \qquad \dfrac{\dfrac{\begin{array}{c}\vdots\\ \llbracket\hat{\uparrow}X\rrbracket \Vdash \llbracket\Psi'\rrbracket\end{array}}{\hat{\uparrow}\llbracket X\rrbracket \Vdash \llbracket\Psi'\rrbracket}\,\text{Lemma 31}}{\llbracket X\rrbracket \vdash \check{\downarrow}\llbracket\Psi'\rrbracket}\,\hat{\uparrow}\dashv\check{\downarrow}}{\dfrac{X' \vdash \check{\downarrow}\llbracket\Psi'\rrbracket}{\hat{\uparrow}X' \Vdash \llbracket\Psi'\rrbracket}\,\hat{\uparrow}\dashv\check{\downarrow}}\,(\mathrm{IH})_X$$

$$\begin{array}{c}\vdots\,\pi\\ \llbracket\Psi\rrbracket\ t\ \llbracket\Phi\rrbracket\end{array} \qquad \begin{array}{c}\vdots\,\pi[\llbracket\Psi'\rrbracket/\llbracket\Phi\rrbracket]\\ \llbracket\Psi\rrbracket\ tt'\ \llbracket\Psi'\rrbracket\end{array}$$

The other cases are treated similarly. ◂

Complements for the proof of completeness (theorem 30). We prove that the equivalence relations $\approx_s$ are congruences, i.e. that they respect the operations and the orders. It will follow that the operations and weakening relations defined on these equivalence classes are well-defined. We only detail the case of $\otimes$ with one pure and one shifted premise, and $\preceq$, the rest being similar.

$$\dfrac{\dfrac{\dfrac{X \approx_{\mathbb{P}} Y}{\llbracket X\rrbracket \vdash \llbracket Y\rrbracket} \qquad \dfrac{\dot{X} \approx_{\mathbb{P}} \dot{Y}}{\llbracket\dot{X}\rrbracket \Vdash \llbracket\dot{Y}\rrbracket}}{\llbracket X\rrbracket \hat{\otimes} \llbracket\dot{X}\rrbracket \vdash \llbracket Y\rrbracket \otimes \llbracket\dot{Y}\rrbracket}\otimes_R \qquad \dfrac{\dfrac{X \approx_{\mathbb{P}} Y}{\llbracket Y\rrbracket \vdash \llbracket X\rrbracket} \qquad \dfrac{\dot{X} \approx_{\mathbb{P}} \dot{Y}}{\llbracket\dot{Y}\rrbracket \Vdash \llbracket\dot{X}\rrbracket}}{\llbracket Y\rrbracket \hat{\otimes} \llbracket\dot{Y}\rrbracket \vdash \llbracket X\rrbracket \otimes \llbracket\dot{X}\rrbracket}\otimes_R}{\dfrac{\text{Lemma 31}\quad \llbracket X\rrbracket \hat{\otimes} \llbracket\dot{X}\rrbracket \vdash \llbracket Y\rrbracket \otimes \llbracket\dot{Y}\rrbracket \qquad \llbracket Y\rrbracket \hat{\otimes} \llbracket\dot{Y}\rrbracket \vdash \llbracket X\rrbracket \otimes \llbracket\dot{X}\rrbracket\quad \text{Lemma 31}}{\llbracket X\hat{\otimes}\dot{X}\rrbracket \vdash \llbracket Y\hat{\otimes}\dot{Y}\rrbracket}}{X\hat{\otimes}\dot{X} \approx_{\mathbb{P}} Y\hat{\otimes}\dot{Y}}$$

$$\dfrac{\dfrac{X \approx_{\mathbb{P}} Y}{\llbracket X\rrbracket \vdash \llbracket Y\rrbracket} \qquad \dfrac{\dfrac{\llbracket Y\rrbracket \vdash \llbracket\Gamma\rrbracket \qquad \dfrac{\Delta \approx_{\mathbb{N}} \Gamma}{\llbracket\Gamma\rrbracket \vdash \llbracket\Delta\rrbracket}}{\llbracket Y\rrbracket \vdash \llbracket\Delta\rrbracket}\,\text{Lemma 34}}{\llbracket X\rrbracket \vdash \llbracket\Delta\rrbracket}\,\text{Lemma 34}$$

For every homogeneous turnstile t, reflexivity,[4] transitivity and antisymmetry of $t^{\mathbb{A}}$ are a consequent of lemma 33, lemma 34 and definition of $\approx_s$ respectively. For heterogeneous turnstiles t, the weakening property of $t^{\mathbb{A}}$ is a consequence of lemma 34. The property (4) of definition 7 is due to to rules $\hat{\uparrow} \dashv \check{\downarrow}$ and $\hat{\uparrow} \dashv \check{\downarrow}$ and (5) to rules $\check{\downarrow}$ and $\hat{\uparrow}$. The adjunction (6) and (7) straightforwardly hold thanks to the corresponding rules in (13). We only develop the example of property $\uparrow^{\mathbb{A}} \dashv \downarrow^{\mathbb{A}}$:

$$\dfrac{\dfrac{\dfrac{\dfrac{\uparrow^{\mathbb{A}}[X]_{\approx_{\mathbb{P}}} \Vdash^{\mathbb{A}} [\dot{\Delta}]_{\approx_{\mathbb{N}}}}{\llbracket\hat{\uparrow}X\rrbracket \Vdash \llbracket\dot{\Delta}\rrbracket}\,\text{Lemma 31}}{\hat{\uparrow}\llbracket X\rrbracket \Vdash \llbracket\dot{\Delta}\rrbracket}\,\hat{\uparrow}\dashv\check{\downarrow}}{\llbracket X\rrbracket \vdash \check{\downarrow}\llbracket\dot{\Delta}\rrbracket}\,\text{Lemma 31}}{[X]_{\approx_{\mathbb{P}}} \vdash^{\mathbb{A}} \downarrow^{\mathbb{A}}[\dot{\Delta}]_{\approx_{\mathbb{N}}}}$$

◂

B Symmetries

Lambek-Grishin calculus exhibits two main symmetries [23]: an order-preserving left-right symmetry $\cdot^{\bowtie}$ and an order-reversing dual symmetry $\cdot^{\infty}$ represented in (18)[5]. We extend them to $\mathbb{FP}.\mathbb{LG}$ and

[4] Reflexivity on structures Ψ such that $\llbracket\Psi\rrbracket\ t\ \llbracket\Psi\rrbracket$ is not defined is explicitly added.
[5] These definitions should be understood as $(A \otimes B)^{\bowtie} = B^{\bowtie} \otimes A^{\bowtie}$, $(A \otimes B)^{\infty} = B^{\infty} \oplus A^{\infty}$, etc.

fD.LG by (19). The dual of a turnstile t is given by (19) through the turnstile interpretation of (15): $t^{\infty} = (t^{\mathbb{A}})^{\infty}$, e.g. $\vdash^{\infty} = \vdash$.

$$\bowtie \quad \frac{A \setminus C \quad A \otimes B \quad A \oplus B \quad C \oslash B}{C / A \quad B \otimes A \quad B \oplus A \quad B \oslash C} \qquad \infty \quad \frac{A \setminus C \quad A \otimes B \quad C / B}{C \oslash A \quad B \oplus A \quad B \oslash C} \tag{18}$$

$$\bowtie \quad \frac{\mathring{M} \setminus_{\ell} \mathring{N} \quad \mathring{Q} \setminus_{r} \mathring{P} \quad \mathring{N} \otimes_{\ell} \mathring{P} \quad \mathring{N} \oplus_{r} \mathring{P} \quad \mathring{N} \oslash_{\ell} \mathring{M} \quad \mathring{P} \oslash_{r} \mathring{Q}}{\mathring{N} /_{r} \mathring{M} \quad \mathring{P} /_{\ell} \mathring{Q} \quad \mathring{P} \otimes_{r} \mathring{N} \quad \mathring{P} \oplus_{\ell} \mathring{N} \quad \mathring{M} \oslash_{r} \mathring{N} \quad \mathring{Q} \oslash_{\ell} \mathring{P}}$$

$$\infty \quad \frac{A \setminus_{\ell} C \quad A \setminus_{r} C \quad A \otimes_{\ell} B \quad A \otimes_{r} B \quad C /_{\ell} B \quad C /_{r} B \quad \leq \quad \preceq \quad \trianglelefteq \quad \succeq \quad \preceq}{C \oslash_{r} A \quad C \oslash_{\ell} A \quad B \oplus_{r} A \quad B \oplus_{\ell} A \quad B \oslash_{r} C \quad B \oslash_{\ell} C \quad \leq \quad \trianglelefteq \quad \trianglelefteq \quad \trianglerighteq \quad \preceq} \tag{19}$$

The presentation of **fD.LG** rules in section 3.1 also reflects the dual symmetry. In equation (12) the dual of rule R is the one displayed on the opposite side of the page w.r.t. the vertical axis. Therefore we have the following property.

▶ **Proposition 35.** *If $\Phi \; t \; \Psi$ is a derivable sequent, then $\Phi^{\bowtie} \; t \; \Psi^{\bowtie}$ and $\Psi^{\infty} \; t^{\infty} \; \Phi^{\infty}$ are also derivable.*

C Examples

In the tradition of parsing-as-deduction [19, 20], various extensions of the Lambek calculus have been proposed to recognize whether sentences are syntactically well-formed and tell apart different readings [24]. A well-formed sentence like (1-a), for example, is semantically ambiguous as shown by the paraphrases (1-b) or (1-c).

(1) a. Everyone seeks some unicorn.
b. For everyone, there is a unicorn such that she seeks it. (∀∃-reading)
c. There is a unicorn such that everyone seeks it. (∃∀-reading)

In particular, the previous readings can be already captured by genuinely different focused proofs of the minimal Lambek logic. In figure 5 and figure 4 we provide **fD.LG**-derivations of the two readings, where every occurrence of atoms n (common nouns) and np (noun phrases) is assigned a positive polarity, and every occurrence of atoms s (sentence) is assigned a negative polarity (cfr. derivation (35) in [24] where the polarity assignment is called 'bias'). Notice how in figure 5 first `some unicorn` is attacked, and only then `everyone` is attacked. In figure 4 is the other way around. Figure 3 shows the signed generation of the end-sequent.

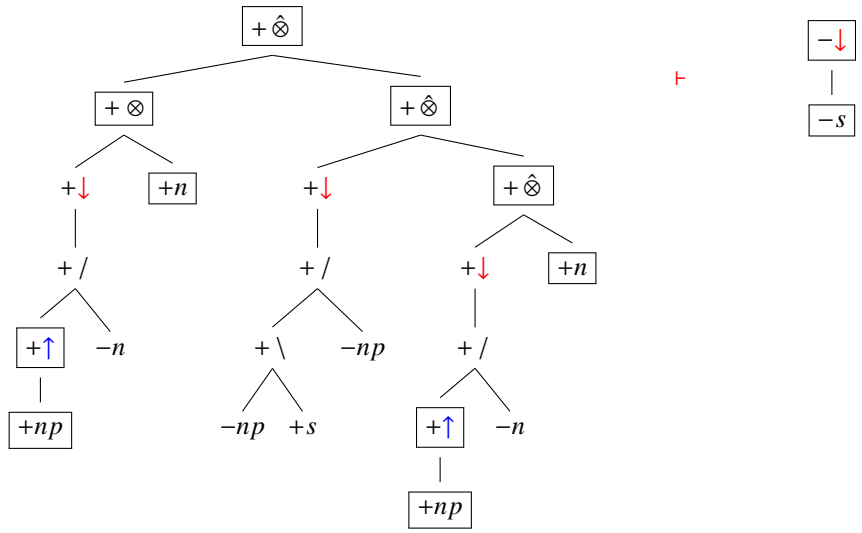


Figure 3 Signed generation tree of the end-sequent in figure 4 and 5. Skeleton nodes are encapsulated in a box where PIA nodes are not.

$$
\dfrac{
\dfrac{
\dfrac{

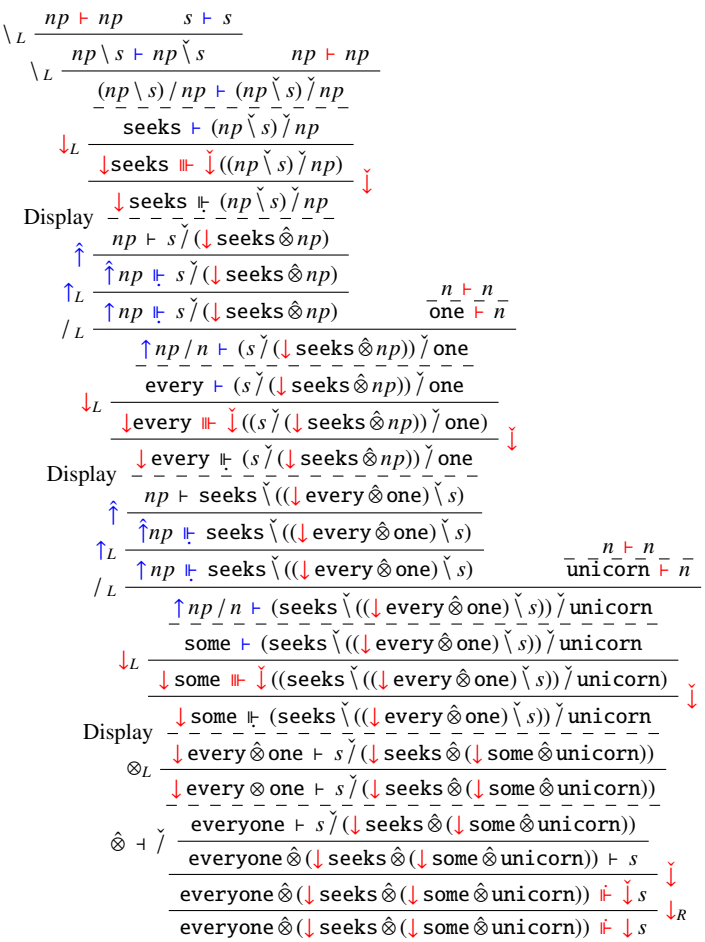


Figure 5 Derivation attached to (1-c): ∃∀-reading.

III Conclusion and future prospects

We showed that the Lambek-Grishin calculus can be presented as a proper Display sequent calculus where every cut-free and ℓ, r-variant-free derivation is strongly focalized, and that this calculus is sound and complete w.r.t. to an algebraic semantics. The many properties of **fD.LG** ranks it as a great candidate for natural language parsing given that proof-search is simplified. Moreover, the identity-of-proof property enables us to identify different structural readings of an ambiguous sentence by unique normal proofs. This makes a natural-language-semantic interpretation possible, like in [24]. Applications to other natural-language-semantic fields, like vector-based models of meaning, would need to extend the algebraic semantics to a categorical semantics. This may be investigating in a future paper.

However, we already have some little answer for the original problem of my internship: how to avoid collapsing of positive and negative operations. In a $\mathbb{FP}.\mathbb{LG}$, contrary to a LG-algebra, it turns out that $\mathbb{N}$ and $\mathbb{P}$ may be non-(dually-)isomorphic. This give more freedom in the finding of an actual practical model where the tensor $\otimes$ and the cotensor $\oplus$ perform different operations. Moreover, appendix A can also lead to a differentiation between two behaviours of $\otimes$, what may have linguistic accuracy. Semantic focalization may be the interest of some future project.

The extension of the class of multi-type proper display calculi to include heterogeneous sequents is a completely written theorem and is planned as a subsequent paper together with Giuseppe (and maybe other colleagues). Regarding the unused graphical tool for sequent calculi (sequent multi-graphs), I plan to investigate it further alone or with another supervisor. However, I still do not know whether this will be part of a PhD thesis or a side project.

A Semantic focalization

▶ Remark 36. We share here an important observation about a property that emerges from fully polarization: every LG-connective could be split into two different ones having different behaviours. In order to be more concrete, let us take the example of $\otimes$ and suppose here we have commutativity of $\otimes$ as a structural rule:

$$\frac{\mathring{X} \hat{\otimes} \mathring{Y} \vdash \mathring{\Delta}}{\mathring{Y} \hat{\otimes} \mathring{X} \vdash \mathring{\Delta}}\ \text{Comm}$$

Fix $p, q \in \mathsf{AtProp}^+$. Because Comm is only applicable in non-focused phase, focalization forbids $q \hat{\otimes} p \vdash p \otimes q$ to be derivable in **fD.LG** +Comm. Yet in most semantics, Comm would be sound by letting $\otimes^{\mathbb{A}}$ be commutative, so that $q^{\mathbb{A}} \otimes^{\mathbb{A}} p^{\mathbb{A}} \leq p^{\mathbb{A}} \otimes^{\mathbb{A}} q^{\mathbb{A}}$ would hold in every model. Therefore, if **fD.LG** +Comm were complete w.r.t. that semantics, strong focalization of cut-free and variant-free **fD.LG** +Comm would fail. Therefore, it is essential to avoid commutativity of $\otimes^{\mathbb{A}}$ in $\mathbb{P}$, but still allowing commutativity of $\otimes^{\mathbb{A}}$ *from the point of view of* $\mathring{\leq}$. The following general definition brings a useful tool to understand how this is possible.

▶ Definition 37. *Given a weakening relation $\preccurlyeq \subseteq \mathcal{A} \times \mathcal{B}$, the relations $\equiv_{\preccurlyeq}$ on $\mathcal{A}$ and ${}_{\preccurlyeq}\equiv$ on $\mathcal{B}$ defined by* (20) *are equivalence relations.*

$$\begin{array}{l} A \equiv_{\preccurlyeq} A' \text{ iff for every } B \in \mathcal{B}, A \preccurlyeq B \iff A' \preccurlyeq B \\ B \,{}_{\preccurlyeq}\!\equiv B' \text{ iff for every } A \in \mathcal{A}, A \preccurlyeq B \iff A \preccurlyeq B' \end{array} \tag{20}$$

The quotiented sets $\mathcal{A}/\equiv_{\preccurlyeq}$ and $\mathcal{B}/{}_{\preccurlyeq}\!\equiv$ remain orders on which $\preccurlyeq$ is well-defined.

Set a $\mathbb{FP}.\mathbb{LG}$ $\mathbb{A}$. The relation $\equiv_{\mathring{\leq}}$ is the identity on $\mathring{\mathbb{P}}$ (it is the case for every order), but $\equiv_{\mathring{\leq}}$ can be coarser. In other words, there could be $\mathring{P}, \mathring{Q} \in \mathring{\mathbb{P}}$ such that $\mathring{P} \neq \mathring{Q}$ and for all $\mathring{N} \in \mathring{\mathbb{N}}$, $\mathring{P} \mathring{\leq} \mathring{N}$ iff $\mathring{Q} \mathring{\leq} \mathring{N}$. Moreover, both relations are compatible with the operations of $\mathbb{A}$. Therefore, we can split the tensor into $\otimes^{\mathbb{A}}$ defined on $\mathring{\mathbb{P}}/\equiv_{\mathring{\leq}}$ (which is in fact the tensor of $\mathbb{A}$) and $\otimes'^{\mathbb{A}}$ defined on $\mathring{\mathbb{P}}/\equiv_{\mathring{\leq}}$. If we wanted to design a semantics for **fD.LG** +Comm, we would ask $\otimes'^{\mathbb{A}}$ to be commutative, and ask nothing for $\otimes^{\mathbb{A}}$.

▶ **Proposition 38.** *In a $\mathbb{FP}.\mathbb{LG}$, if $\downarrow\uparrow = \mathsf{id}_{\mathbb{P}}$ and $\downarrow\uparrow = \mathsf{id}_{\mathring{\mathbb{P}}}$ then $\equiv_{\mathring{\leq}}$ is the identity $\mathring{\mathbb{P}}$.*

Proof. Assume $P \equiv_{\overline{\preccurlyeq}} Q$. We have by reflexivity and adjunction $P \leq \downarrow\uparrow P$, so $P \,\overline{\preccurlyeq}\, \uparrow P$. By assumption, $Q \,\overline{\preccurlyeq}\, \uparrow P$ also holds, and $Q \leq \downarrow\uparrow P$ too by definition of $\overline{\preccurlyeq}$. Equation $\downarrow\uparrow = \mathsf{id}_{\mathbb{P}}$ gives $Q \leq P$. By symmetry of the problem, $P \leq Q$, so $P = Q$ by antisymmetry of $\leq$.

In the case $\dot{P} \equiv_{\overline{\preccurlyeq}} \dot{Q}$, by a similar reasoning we get we get $\dot{P} = \dot{Q}$ using $\downarrow\uparrow = \mathsf{id}_{\dot{\mathbb{P}}}$. The cases where $\mathring{P}$ and $\mathring{Q}$ do not have the same purity are impossible. ◀

▶ Remark 39. A consequence of remark 36 and proposition 38 is that shifts must not be isomorphisms, at least in some models. Indeed, if shifts are isomorphisms, $\otimes'^{\mathbb{A}}$ collapse to $\otimes^{\mathbb{A}}$ because $\equiv_{\mathring{\leq}}$ is the identity of $\mathring{\mathbb{P}}$, and semantic focalization is lost.

ℓ, r-variants are subtle to handle. The way they work in definition 7 is the only way we found to prevent us to require isomorphicity of shifts. In particular, we consider it safer to avoid defining ℓ, r-variants using shifts (e.g. avoiding $N \otimes^{\mathbb{A}}_{\ell} P$ to be equal to $\uparrow^{\mathbb{A}}(\downarrow^{\mathbb{A}} N \otimes^{\mathbb{A}} P)$).